

SOVEREIGN-GRADE STANDARD | GRAND MASTER DOCUMENT

MERLIN

Masyarakat Ekosistem Rumput Laut Indonesia



MERLIN 2030

**MASTER DOCUMENT 165 PGA
COMPLETED**

Grand Design Ekosistem Rumput Laut Indonesia

20 Domain × 165 Dokumen — Arsitektur Lengkap Hilirisasi Rumput Laut Berdaulat

Versi	1.0.0
Tanggal Terbit	Juli 2026
Status	Final — Ready for Implementation
Klasifikasi	Strategis — Untuk Pemerintah, Investor & Mitra

DISEUSUN OLEH DEWAN PAKAR 46 LINTAS BIDANG • 150+ TAHUN PENGALAMAN

EXECUTIVE SUMMARY

MERLIN (Masyarakat Ekosistem Rumput Laut Indonesia) adalah inisiatif strategis nasional yang bertujuan menjadikan Indonesia sebagai Pusat Industri Hilirisasi Rumput Laut Dunia pada tahun 2030. Dokumen Master 165 PGA (Policy, Governance & Architecture) Completed ini merupakan arsitektur dokumentasi paling lengkap, terintegrasi, dan berkelas sovereign yang pernah disusun untuk ekosistem maritim Indonesia, melampaui standar McKinsey, BCG, Bain, dan lembaga internasional seperti UN, WTO, dan World Bank.

Dokumen ini disusun oleh Dewan Pakar 46 lintas bidang dengan total pengalaman gabungan lebih dari 150 tahun, mencakup seluruh aspek ekosistem rumput laut dari hulu (budidaya) hingga hilir (100+ produk turunan), dari tata kelola hingga diplomasi internasional, dari keuangan syariah hingga keadilan sosial, dan dari teknologi digital hingga peradaban masa depan. Total 20 Domain strategis dijabarkan ke dalam 165 dokumen PGA yang masing-masing memiliki tujuan, elemen kunci, pemilik (owner), dan metrik keberhasilan yang jelas.

Highlight Strategis MERLIN 2030

Investasi Total	Rp680 Triliun
Ekspor Tahunan (2030)	Rp500 Triliun
Eco Blue Industrial Park	40 Kawasan Zero Waste
Blue Tourism Destination	40 Kawasan Kelas Dunia
Lapangan Kerja Baru	5 Juta Tenaga Kerja
Blue Carbon Sequestration	12 Juta Ton CO ₂ /Tahun
Produk Hilir	100+ Produk Terdiversifikasi
Asosiasi Terpadu	10 Asosiasi MERLIN
Petani Terlibat	1,5 Juta KK sejahtera
Dokumen PGA	165 Dokumen Lengkap

Struktur Dokumen

Dokumen ini terdiri dari 20 Domain yang mencerminkan tata kelola lengkap suatu ekosistem ekonomi berdaulat: (1) Identity & Civilization, (2) Strategy & Direction, (3) Governance & Legal, (4) Product/Service/Solution, (5) Operations & Delivery, (6) Finance, Risk & Control, (7) People, Culture & Leadership, (8) Brand, Growth & Stakeholders, (9) Data, Intelligence & Legacy, (10) Diplomacy & International Relations, (11) Sovereignty Economics & Islamic Finance, (12) Social Justice & Human Rights, (13) Future Civilization & Frontier Science, (14) Defense, Security & Resilience, (15)

Technology & Digital Sovereignty, (16) Media, Narrative & Communications, (17) Education, Research & Knowledge, (18) Healthcare, Wellness & Social, (19) Infrastructure & Urban Planning, dan (20) Legal, Constitutional & Justice System.

Setiap dokumen PGA disusun dengan struktur baku: ID unik (PGA-001 s/d PGA-165), judul dokumen, tujuan (purpose), elemen-elemen kunci (key elements), dan pemilik tanggung jawab (owner). Pendekatan ini memastikan akuntabilitas, traceability, dan kemudahan implementasi di seluruh tingkat organisasi MERLIN.

DAFTAR ISI

Bagian I — Executive Summary

Bagian II — Daftar Isi

Bagian III — 165 PGA Documents (20 Domain)

RINCIAN 20 DOMAIN × 165 DOKUMEN PGA

Domain	Nama Domain	Range PGA	Jumlah
1	IDENTITY & CIVILIZATION	PGA-01 s/d PGA-08	8
2	STRATEGY & DIRECTION	PGA-09 s/d PGA-16	8
3	GOVERNANCE & LEGAL	PGA-17 s/d PGA-24	8
4	PRODUCT / SERVICE / SOLUTION	PGA-25 s/d PGA-32	8
5	OPERATIONS & DELIVERY	PGA-33 s/d PGA-40	8
6	FINANCE, RISK & CONTROL	PGA-41 s/d PGA-48	8
7	PEOPLE, CULTURE & LEADERSHIP	PGA-49 s/d PGA-56	8
8	BRAND, GROWTH & STAKEHOLDERS	PGA-57 s/d PGA-64	8
9	DATA, INTELLIGENCE & LEGACY	PGA-65 s/d PGA-72	8
10	DIPLOMACY & INTERNATIONAL RELATIONS	PGA-73 s/d PGA-80	8
11	SOVEREIGNTY ECONOMICS & ISLAMIC FINANCE	PGA-81 s/d PGA-88	8

12	SOCIAL JUSTICE, EQUITY & HUMAN RIGHTS	PGA-89 s/d PGA-96	8
13	FUTURE CIVILIZATION & FRONTIER SCIENCE	PGA-97 s/d PGA-108 (12 dokumen)	12
14	DEFENSE, SECURITY & RESILIENCE	PGA-109 s/d PGA-116	8
15	TECHNOLOGY, INNOVATION & DIGITAL SOVEREIGNTY	PGA-117 s/d PGA-124	8
16	MEDIA, NARRATIVE & COMMUNICATIONS POWER	PGA-125 s/d PGA-132	8
17	EDUCATION, RESEARCH & KNOWLEDGE PRODUCTION	PGA-133 s/d PGA-140	8
18	HEALTHCARE, WELLNESS & SOCIAL SERVICES	PGA-141 s/d PGA-148	8
19	INFRASTRUCTURE, URBAN PLANNING & DEVELOPMENT	PGA-149 s/d PGA-156	8
20	LEGAL, CONSTITUTIONAL & JUSTICE SYSTEM	PGA-157 s/d PGA-165 (9 dokumen)	9

TOTAL: 20 DOMAIN × 165 DOKUMEN PGA

DOMAIN 1 | IDENTITY & CIVILIZATION

Fondasi Jati Diri MERLIN — Siapa Kita, Dari Mana, Ke Mana

Range: PGA-01 s/d PGA-08 | Total: 8 Dokumen

PGA-01: Vision Statement MERLIN 2030

Tujuan: Menetapkan visi tunggal MERLIN sebagai panduan tertinggi seluruh ekosistem rumput laut Indonesia menuju 2030 dan beyond.

Elemen Kunci:

- Visi Utama: "Menjadikan Indonesia sebagai Pusat Industri Hilirisasi Rumput Laut Dunia pada 2030, Berkelanjutan, Berdaulat, dan Bermartabat."
- Visi Paragraf: MERLIN hadir sebagai ekosistem terpadu dari hulu hingga hilir yang mengubah 7,1 juta hektar potensi budidaya rumput laut nasional menjadi mesin ekonomi biru senilai Rp500 triliun/tahun, mengangkat 5 juta petani dan nelayan pesisir dari kemiskinan struktural, serta menempatkan Indonesia sebagai OPEC Biru dunia untuk karbon laut dan biomaterial.
- Dampak Visi Tercapai: (1) Pendapatan petani naik 5-8 kali lipat dari Rp1,2 juta menjadi Rp6-9 juta/bulan; (2) Devisa ekspor Rp500 triliun/tahun dari produk hilir; (3) 5 juta lapangan kerja baru; (4) 40 Eco Blue Industrial Park beroperasi; (5) 12 juta ton CO2 terserap/tahun melalui blue carbon.
- Prinsip Visi (Non-Negotiable): Berdaulat (sumber daya dikuasai rakyat), Berkelanjutan (zero waste, blue carbon positive), Berkeadilan (petani sejahtera, bukan hanya korporasi), Berdaya Saing Global (kualitas internasional), Berilmu Pengetahuan (riset-driven).
- Koneksi ke 16 Hak Member MERLIN: Visi menjamin setiap member mendapatkan harga jaminan, asuransi panen, akses pendidikan, pelayanan kesehatan, jaminan hari tua, rumah layak, pendidikan anak, dan 9 hak fundamental lainnya.
- Antitesis Visi: MERLIN BUKAN eksploitasi tambang laut, BUKAN monokultur industri, BUKAN ekspor bahan mentah, BUKAN alat asing menguasai pesisir, BUKAN greenwashing korporasi.
- Metrik Visi: Produksi 25 juta ton/tahun, Ekspor Rp500 triliun, 40 EBIP, 5 juta KK, 12 juta ton CO2, 100+ produk hilir, 1000+ paten.
- Visi per Tier: (1) Petani = sejahtera; (2) Koperasi = mandiri; (3) Pabrik = world-class; (4) Asosiasi = berdaya; (5) Pemerintah = berdaulat; (6) Dunia = bermitra.
- Visi Inter-Temporal: 2026 fondasi legal, 2028 20 EBIP, 2030 40 EBIP pusat dunia, 2035 biomaterial global leader, 2040 blue carbon superpower, 2050 maritime civilization reborn.
- Visi Singkat: (5 kata) Indonesia Pusat Rumput Laut Dunia; (10 kata) Hilirisasi Rumput Laut Berkelanjutan untuk Indonesia Emas 2030; (15 kata) MERLIN Mengubah Laut Nusantara Menjadi Mesin Ekonomi Biru Berkelanjutan dan Berdaulat; (25 kata) Masyarakat Ekosistem

Rumput Laut Indonesia Mewujudkan Hilirisasi Total, Zero Waste, Blue Carbon, dan Kesejahteraan Petani Pesisir Menuju Indonesia Pusat Industri Rumput Laut Dunia 2030.

Owner: Dewan Pembina MERLIN / Presiden RI

PGA-02: Mission Statement MERLIN

Tujuan: Menjabarkan visi menjadi misi operasional yang dapat dieksekusi oleh 10 asosiasi dan seluruh member.

Elemen Kunci:

- Misi Utama: Membangun ekosistem rumput laut nasional terpadu dari hulu hingga hilir yang berkelanjutan, berdaya saing global, dan mengangkat kesejahteraan masyarakat pesisir.
- 7 Misi Prioritas: (1) Membangun ekosistem hulu-hilir terpadu; (2) Meningkatkan kesejahteraan 1,5 juta KK petani; (3) Mengembangkan 100+ produk hilir bernilai tinggi; (4) Membangun 40 Eco Blue Industrial Park; (5) Mengembangkan 40 kawasan wisata bahari kelas dunia; (6) Menjadikan Indonesia pusat inovasi rumput laut; (7) Menguasai pasar global dengan brand MERLIN.
- Misi per Fungsi: Bisnis (efisiensi rantai nilai, profitabilitas, ROI 15%+); Sosial (kesejahteraan, pendidikan, kesehatan member); Kultural (revitalisasi maritime Nusantara, kearifan lokal); Ekologis (zero waste, blue carbon, restorasi laut).
- Metrik Keberhasilan Misi: Produksi 25 juta ton/tahun, ekspor Rp500 triliun, 5 juta lapangan kerja, 40 EBIP operasional, 100+ produk hilir, 12 juta ton CO2 terserap, indeks kesejahteraan petani naik 300%.
- Narasi: MERLIN adalah peradaban baru pesisir Indonesia, di mana laut bukan lagi sumber kemiskinan, melainkan sumber kemakmuran yang abadi.

Owner: Ketua Umum MERLIN

PGA-03: Founders Philosophy & Guiding Principles

Tujuan: Mendokumentasikan filosofi pendiri MERLIN dan 9 pilar prinsip panduan yang mengikat seluruh ekosistem.

Elemen Kunci:

- Filosofi Pendiri: "Laut adalah ibu, rumput laut adalah air susu ibu, dan petani pesisir adalah anak yang berhak menyusu." MERLIN lahir dari keprihatinan 1,5 juta KK petani rumput laut yang miskin di tengah kekayaan laut terbesar kedua dunia.
- 9 Pilar Prinsip Panduan: (1) Berdaulat atas sumber daya laut; (2) Berkelanjutan secara ekologis; (3) Berkeadilan bagi petani; (4) Berilmu pengetahuan dan riset; (5) Bersatu dalam 10 asosiasi; (6) Berdaya saing global; (7) Berorientasi hilirisasi total; (8) Berkomitmen zero waste; (9) Berkhidmat pada bangsa dan agama.
- Latar Belakang Filosofis: Terinspirasi dari Jalur Rempah Nusantara, Bandar Laut Majapahit, VOC yang menguasai perdagangan rempah, dan konsep Bahari Nusantara Ki Hajar Dewantara.
- Pengaruh Historis: ARLI (1989), ASTRULI, ASPPERLI sebagai asosiasi eksisting; CEDUS-UNHAS sebagai pusat riset; Koperasi Mina Agar Makmur sebagai model koperasi sukses.

- Implementasi: 9 pilar tertulis dalam AD/ART, diucapkan dalam ikrar member, diajarkan dalam akademi MERLIN, dan diukur dalam KPI tahunan.
- Warisan Filosofis: MERLIN sebagai gerakan kemandirian ekonomi maritim Indonesia abad ke-21, melanjutkan semangat Sumpah Pemuda 1928 dan Proklamasi 1945.

Owner: Dewan Pakar & Pendiri MERLIN

PGA-04: Story of Origin / Narrative of Origin

Tujuan: Mendokumentasikan kisah pendirian MERLIN sebagai narasi inspiratif yang menyatukan seluruh stakeholder.

Elemen Kunci:

- Kisah Pendirian: MERLIN lahir dari pertemuan 46 dewan pakar lintas bidang di Jakarta awal 2026 yang menyimpulkan bahwa Indonesia menguasai 8,6% produksi rumput laut dunia namun hanya menikmati 2,1% nilai ekonominya karena ekspor bahan mentah.
- Konteks Historis: Indonesia memiliki garis pantai 108.000 km (terpanjang ke-2 dunia), 7,1 juta hektar potensi budidaya rumput laut, namun 1,5 juta KK petani hidup di bawah garis kemiskinan dengan rata-rata pendapatan Rp1,2 juta/bulan.
- Tokoh Kunci: Pendiri dari ARLI (Safari Azis), CEDUS-UNHAS, APINDO, KADIN, BRIN, serta 46 dewan pakar lintas bidang dengan total 150+ tahun pengalaman.
- Titik Balik: Domain merlin.com dibeli Merlin Entertainments UK, sehingga dipilih domain merlin.blue sebagai brand global dengan filosofi ekonomi biru + laut + karbon.
- Narasi Inspiratif: "Dari 1,5 juta petani yang miskin, lahir 1,5 juta keluarga yang sejahtera. Dari ekspor rumput kering Rp15 ribu/kg, lahir ekspor karagenan Rp450 ribu/kg. Dari laut yang terlupakan, lahir peradaban bahari baru."
- Legenda: MERLIN diambil dari nama penyihir legendaris yang merekayasa peradaban, melambangkan transformasi magis ekosistem rumput laut menjadi mesin ekonomi.

Owner: Sekretariat MERLIN

PGA-05: Core Values Charter (9 Pilar MERLIN)

Tujuan: Menetapkan 9 nilai inti yang menjadi kompas moral dan operasional seluruh ekosistem MERLIN.

Elemen Kunci:

- Pilar 1 - AMANAH: Kepercayaan rakyat dan negara dijaga dengan integritas total. Setiap rupiah dana publik transparan, setiap keputusan dapat diaudit.
- Pilar 2 - KEADILAN: Petani mendapat harga jaminan Rp18.000/kg, bukan Rp6.000/kg. Kesejahteraan dibagi proporsional dari hulu ke hilir dengan rasio 40:30:30.
- Pilar 3 - TRANSPARANSI: Seluruh data produksi, harga, keuangan, dan keputusan publik melalui dashboard real-time yang dapat diakses member.

- Pilar 4 - KEMASLAHATAN: Setiap keputusan harus memberi manfaat nyata kepada masyarakat pesisir, bukan hanya shareholder.
- Pilar 5 - KESEDERHANAAN: Operasional efisien, biaya overhead <8% pendapatan, fokus pada value creation bukan vanity.
- Pilar 6 - KEBERSAMAAN: 10 asosiasi bersatu dalam 1 rumah MERLIN, keputusan melalui musyawarah, tidak ada asosiasi yang mendominasi.
- Pilar 7 - KELESTARIAN: Zero waste, blue carbon positive, restorasi ekosistem laut, 0% limbah keluar kawasan industri.
- Pilar 8 - KEILMUAN: Riset-driven, semua praktik berbasis sains, 5% pendapatan untuk R&D, kemitraan dengan 50+ universitas.
- Pilar 9 - KEDAULATAN: Sumber daya laut dikuasai rakyat Indonesia, ekspor produk hilir bukan bahan mentah, brand MERLIN milik bangsa.
- Implementasi: Setiap pilar memiliki KPI, audit tahunan, dan sanksi bagi pelanggar. Ikrar 9 pilar diucapkan setiap member saat bergabung.
- Metrik Kepatuhan: Survei tahunan ke seluruh member, skor minimum 85% untuk mempertahankan sertifikasi MERLIN.

Owner: Dewan Pengawas & Kehormatan MERLIN

PGA-06: Identity & Positioning Paper

Tujuan: Menetapkan identitas brand MERLIN dan positioning strategis di pasar global.

Elemen Kunci:

- Brand Identity: MERLIN = Masyarakat Ekosistem Rumput Laut Indonesia. Logo berbentuk gelombang laut + rumput laut + bintang (0% waste cycle). Warna utama deep ocean blue (#0A4D8C), seaweed green (#007A5A), gold (#C9A227).
- Market Positioning: "The World's First Sovereign-Grade Seaweed Ecosystem" - posisi premium sebagai ekosistem rumput laut terlengkap dan paling berkelanjutan di dunia.
- Diferensiasi Kompetitif: (1) 10 asosiasi terpadu (pesaing hanya 1-2 asosiasi); (2) 40 Eco Blue Industrial Park (pesaing belum ada); (3) Zero Waste certified (pesaing belum bersertifikat); (4) Blue Carbon monetized (pesaing baru pilot); (5) 100+ produk hilir (pesaing 10-20 produk).
- Persona Organisasi: Bijaksana (sage), Berdaulat (sovereign), Berkelanjutan (sustainable), Berdaya (empowering), Berkelas (premium).
- Voice & Tone: Formal-profesional untuk dokumen resmi, inspiratif-naratif untuk publik, teknis-presisi untuk industri, hangat-empatis untuk petani.
- Visual Identity Guidelines: Logo minimum size 2cm, clear space 1x height logo, typography utama Arial Bold (heading) dan Times New Roman (body), fotografi laut biru cerah dan petani tersenyum.

Owner: Direktorat Brand & Komunikasi MERLIN

PGA-07: Manifesto / Declaration of Purpose

Tujuan: Deklarasi formal MERLIN kepada bangsa dan dunia tentang tujuan, komitmen, dan ajakan kolaborasi.

Elemen Kunci:

- Pernyataan Manifesto: "Kami, Masyarakat Ekosistem Rumput Laut Indonesia, dengan ini mendeklarasikan komitmen suci untuk mengubah lautan Nusantara dari sumber kemiskinan menjadi sumber kemakmuran abadi, dari ekspor bahan mentah menjadi pusat industri hilir dunia, dari ketergantungan asing menjadi kedaulatan ekonomi biru."
- Komitmen Organisasi: (1) Harga jaminan Rp18.000/kg untuk petani; (2) Zero waste di 40 EBIP; (3) 5 juta lapangan kerja 2030; (4) Rp500 triliun ekspor 2030; (5) 12 juta ton CO2 terserap/tahun.
- Ajakan Stakeholder: Kepada pemerintah (UU Rumput Laut), investor (Rp680 triliun opportunity), akademisi (50+ universitas), petani (bergabung koperasi), dunia (offtake produk MERLIN).
- Narasi: Ini bukan sekadar bisnis. Ini adalah gerakan kemandirian ekonomi maritim Indonesia. Ini adalah pembuktian bahwa bangsa bahari mampu menguasai ekonomi lautnya sendiri.
- Call to Action: Bergabunglah dengan MERLIN. Belilah produk MERLIN. Dukunglah UU Rumput Laut. Investasikan di Eco Blue Industrial Park. Jadilah bagian dari sejarah.

Owner: Dewan Pembina MERLIN

PGA-08: Whitepaper / Concept Paper MERLIN 2030

Tujuan: Dokumen konsep utama yang merangkum seluruh ekosistem MERLIN untuk audiens eksekutif dan investor.

Elemen Kunci:

- Executive Summary: MERLIN adalah ekosistem terpadu rumput laut Indonesia senilai Rp680 triliun investasi, menghasilkan Rp500 triliun ekspor/tahun, 5 juta lapangan kerja, 40 Eco Blue Industrial Park, dan 40 Blue Tourism destination pada 2030.
- Problem Statement: Indonesia produksi 8,6% dunia tetapi nilai ekonomi hanya 2,1%. Petani miskin (Rp1,2 juta/bulan). Hilirisasi rendah. Belum ada kawasan industri rumput laut terpadu. Belum ada roadmap nasional.
- Solusi MERLIN: 10 asosiasi terpadu, 40 EBIP zero waste, 40 Blue Tourism, harga jaminan petani, blue carbon monetized, 100+ produk hilir, brand global MERLIN.
- Market Opportunity: Pasar rumput laut global USD 22 billion (2025) → USD 45 billion (2035). Indonesia target kuota 25% = USD 11 billion/tahun.
- Business Model: Hulu (budidaya 25 juta ton) → Koperasi (collecting) → EBIP (olah 100+ produk) → Blue Tourism (pariwisata) → Ekspor global. Zero waste, blue carbon sebagai revenue stream tambahan.
- Go-to-Market: Fase 1 (2026) akuisisi 3 asosiasi eksisting; Fase 2 (2027) 10 EBIP pilot; Fase 3 (2028-2030) 40 EBIP scale-up global.
- Financial Projections: Investasi Rp680 triliun, Revenue 2030 Rp850 triliun, EBITDA margin 25%, payback period 7 tahun, IRR 18%.

- Team & Governance: Dewan Pembina (Presiden RI), Dewan Pakar (46 lintas bidang), Dewan Pengawas, Ketua Umum, 10 Ketua Asosiasi, CEO MERLIN Holding.
- Roadmap & Milestones: 2026 legal-formasi, 2027 10 EBIP, 2028 20 EBIP + Blue Tourism, 2029 30 EBIP + ekspor melonjak, 2030 40 EBIP pusat dunia.

Owner: CEO MERLIN Holding

◆ Akhir Domain 1 — 8 Dokumen PGA ◆

DOMAIN 2 | STRATEGY & DIRECTION

Arah Strategis MERLIN — Ke Mana Bergerak dan Bagaimana Menang

Range: PGA-09 s/d PGA-16 | Total: 8 Dokumen

PGA-09: Strategic Masterplan 3-10 Tahun MERLIN

Tujuan: Masterplan strategis MERLIN 2026-2035 dengan analisis SWOT/PESTEL lengkap dan inisiatif strategis terprioritas.

Elemen Kunci:

- Executive Summary: Masterplan 10 tahun MERLIN dengan investasi Rp680 triliun menghasilkan 40 EBIP, 40 Blue Tourism, 5 juta lapangan kerja, dan Rp500 triliun ekspor/tahun pada 2030.
- Situational Analysis SWOT: Strengths (garis pantai 108.000 km, 7,1 juta ha potensi, 1,5 juta KK petani, ARLI/Astruli eksisting); Weaknesses (petani miskin, hilirisasi rendah, logistik mahal, belum ada kawasan industri); Opportunities (pasar global USD 45 billion 2035, blue carbon trend, bioplastic demand, ESG investing); Threats (iklim, kompetitor Filipina/China, regulasi, pest/disease).
- PESTEL: Political (UU Rumput Laut dalam proses), Economic (blue economy RPJMN), Social (1,5 juta KK petani), Technological (biotech, MRV carbon), Environmental (climate change, ocean acidification), Legal (Perpres Blue Economy).
- Strategic Objectives 2026-2035: (1) 40 EBIP 2030; (2) 5 juta KK sejahtera; (3) Rp500 triliun ekspor; (4) 12 juta ton CO₂; (5) 1000+ paten; (6) Brand MERLIN global top 3.
- Strategic Initiatives: 10 asosiasi, 40 EBIP, 40 Blue Tourism, Akademi MERLIN, Bursa Carbon Biru, Lab Nasional Rumput Laut, Pelabuhan Ekspor, Brand Global.
- Resource Allocation: Hulu 25%, Hilir 35%, Pariwisata 20%, Logistik 8%, Riset 7%, Brand 5%.
- Timeline & Milestones: 2026 formasi legal + masterplan, 2027 10 EBIP, 2028 20 EBIP, 2029 30 EBIP, 2030 40 EBIP + pusat dunia, 2035 biomaterial global leader.
- Risk Assessment: 12 risiko utama dengan mitigasi (iklim, regulasi, kompetitor, teknologi, sosial, finansial).
- Governance & Oversight: Dewan Pembina quarterly review, Dewan Pakar monthly advisory, audit tahunan independen.

Owner: CEO MERLIN + Dewan Pakar

PGA-10: Business Model & Value Architecture MERLIN

Tujuan: Business model canvas MERLIN dengan arsitektur nilai dari hulu hingga hilir.

Elemen Kunci:

- Business Model Canvas: 9 blok lengkap (Customer Segments, Value Propositions, Channels, Customer Relationships, Revenue Streams, Key Resources, Key Activities, Key Partnerships, Cost Structure).
- Value Proposition: Bagi petani (harga jaminan 3x lipat, asuransi panen, pendidikan anak); Bagi industri (supply bahan baku stabil, kualitas konsisten, volume besar); Bagi investor (IRR 18%, ESG-compliant, brand premium); Bagi dunia (produk berkelanjutan, blue carbon, zero waste).
- Revenue Streams: (1) Ekspor produk hilir 60%; (2) Blue Tourism 15%; (3) Blue Carbon credit 10%; (4) Bioteknologi & paten 8%; (5) Jasa logistik & cold storage 4%; (6) Pendidikan & sertifikasi 3%.
- Cost Structure: Bahan baku 35%, operasional pabrik 22%, logistik 12%, SDM 10%, R&D 5%, marketing 4%, overhead 7%, depreciation 5%.
- Key Partners: KKP, BRIN, 50+ universitas, ARLI/Astruli/Aspperli, Unilever/Loreal/Loccitane (offtaker), Standard Chartered (blended finance), IFC/ADB (investor).
- Key Activities: Budidaya, pengolahan hilir, riset, pariwisata, ekspor, sertifikasi, blue carbon verification.
- Customer Segments: B2B (FMCG, pharma, kosmetik, food), B2G (carbon credit buyer), B2C (wisatawan, consumer products).
- Unit Economics: EBIP average revenue Rp2.5 triliun/tahun, cost Rp1.8 triliun, EBITDA Rp625 miliar, margin 25%, payback 7 tahun.

Owner: CFO + CSO MERLIN

PGA-11: Market Landscape & Industry Map

Tujuan: Pemetaan pasar global rumput laut dan posisi kompetitif Indonesia.

Elemen Kunci:

- Market Size & Growth: Global seaweed market USD 22 billion (2025) → USD 45 billion (2035), CAGR 7.4%. Indonesia produksi 8,6% (3,6 juta ton) tapi nilai ekonomi hanya 2,1%.
- Industry Trends: Hilirisasi (karragenan, agar, alginat), bioplastic (USD 8 billion 2030), blue carbon (USD 50 billion 2030), nutraceutical (CAGR 9%), cosmeceutical (CAGR 8%).
- Competitive Landscape: Top 5 produsen - China (50%), Indonesia (8,6%), Philippines (5,5%), Korea (4,2%), Japan (3,8%). Top processor - CP Kelco, FMC, DuPont, Cargill, Ingredion.
- Regulatory Environment: EU Novel Foods, US FDA GRAS, China GB standard, Halal MUI, organic USDA/EU. Codex Alimentarius for seaweed.
- Technology Landscape: Smart aquaculture (IoT, AI), biorefinery, enzymatic extraction, genetic selection, MRV carbon, blockchain traceability.
- Customer Insights: Unilever butuh 50.000 ton karragenan/tahun, L'Oreal cari alginat premium, Nestle cari nutraceutical, EU cari bioplastic.
- Opportunity Mapping: 7 segmen prioritas - (1) Food ingredient, (2) Pharma excipient, (3) Cosmeceutical, (4) Bioplastic, (5) Bioenergy, (6) Animal feed, (7) Blue carbon.

Owner: CSO MERLIN

PGA-12: Go-To-Market Strategy MERLIN

Tujuan: Strategi masuk pasar untuk setiap segmen dan produk hilir MERLIN.

Elemen Kunci:

- Target Market: Tier 1 (EU, US, Japan, Korea - premium), Tier 2 (China, India, Middle East - volume), Tier 3 (ASEAN, Africa - emerging).
- Value Proposition per Segment: Food (kualitas food-grade, halal, traceable); Pharma (USP/EP grade, GMP); Kosmetik (organic, fair trade); Bioplastic (ISCC+ certified); Blue Carbon (Verra/VCS).
- Pricing Strategy: Premium pricing (15-25% di atas pasar) berdasarkan brand MERLIN, sertifikasi zero waste, dan traceability blockchain.
- Distribution Channels: Direct B2B (key accounts Unilever, L'Oreal), Distributor (regional), E-commerce B2B (Alibaba, Indiamart), Retail B2C (premium stores).
- Sales Strategy: Solution selling (bukan commodity), long-term offtake (5-10 tahun), volume discount, technical support.
- Marketing Strategy: Trade show (Hi Europe, Vitafoods, CPSE), thought leadership (whitepaper, webinar), PR (Bloomberg, Reuters ESG), digital (LinkedIn B2B, content marketing).
- Launch Plan: 2026 soft launch 3 produk pilot, 2027 launch 25 produk di EU, 2028 launch 50 produk global, 2030 launch 100+ produk.
- Metrics & KPIs: Customer acquisition cost <USD 50k, conversion rate 15%, retention 85%, NPS 60+, market share 10% by 2030.

Owner: CMO MERLIN

PGA-13: Ecosystem Strategy MERLIN

Tujuan: Strategi membangun ekosistem 10 asosiasi + 50+ partner strategis sebagai flywheel growth.

Elemen Kunci:

- Ecosystem Map: Pusat MERLIN → 10 Asosiasi (APRLN, AKNBB, AIPRL, ABBR, ABBION, APPOTL, APBB, AKBIN, ARTK, APERL) → 50+ partner (pemerintah, riset, finansial, korporasi, internasional).
- Partner Ecosystem: Pemerintah (KKP, ESDM, Kemenparekraf, KLHK, Bappenas); Riset (BRIN, ITS, UNHAS, IPB, UI); Finansial (Bank Mandiri, BRI, Standard Chartered, IFC); Korporasi (Unilever, L'Oreal, Garuda); Internasional (UNDP, FAO, World Bank).
- Platform Strategy: MERLIN Platform (digital marketplace B2B, traceability blockchain, data dashboard, e-learning).
- Network Effects: Setiap petani baru meningkatkan volume → menarik pabrik → menarik offtaker → menarik investor → menarik lebih banyak petani. Flywheel effect.

- Ecosystem Governance: Dewan Ekosistem quarterly meeting, shared KPIs, profit-sharing mechanism, conflict resolution protocol.
- Monetization Model: Transaction fee 1-2%, subscription SaaS USD 1k-10k/bulan, sertifikasi USD 5k-50k, data licensing.
- Growth Flywheel: Petani → Volume → Pabrik → Produk → Customer → Revenue → Reinvest → Riset → Petani (loop).

Owner: Chief Ecosystem Officer MERLIN

PGA-14: Expansion / International Strategy

Tujuan: Strategi ekspansi MERLIN ke pasar global dan posisi Indonesia sebagai leader.

Elemen Kunci:

- International Market Prioritization: Tier 1 (EU 30%, US 20%, Japan 15%, Korea 10%); Tier 2 (China 10%, Middle East 5%, India 5%); Tier 3 (ASEAN 5%).
- Entry Mode: Export (2030), Joint Venture (2032), Subsidiary (2035), Licensing (2038), Franchise Blue Tourism (2040).
- Localization Framework: Sertifikasi lokal (Halal MUI, EU organic, USDA), bahasa packaging, regulatory compliance, cultural adaptation.
- Partnership Strategy: Distributor eksklusif per region, technical partner untuk aplikasi, R&D partner untuk inovasi.
- Regulatory Compliance: EU REACH, US FDA, China GB, Japan Food Sanitation, Halal MUI, Kosher, organic certifications.
- Risk Management: Currency hedging, political risk insurance (MIGA), supply chain diversification, force majeure protocol.
- Success Metrics: Export USD 30 billion 2030, 50+ countries served, 100+ distributor partner, 5 JV overseas, brand awareness 60% di target market.

Owner: Director International MERLIN

PGA-15: Competitive Moat & Advantage Paper

Tujuan: Analisis moat kompetitif MERLIN yang sulit ditiru kompetitor.

Elemen Kunci:

- Moat 1 - Skala Ekosistem: 10 asosiasi terpadu dengan 1,5 juta KK petani. Kompetitor butuh 10-15 tahun meniru. 40 EBIP zero waste.
- Moat 2 - Brand Sovereign: MERLIN didukung pemerintah Indonesia, brand kedaulatan ekonomi biru. Kompetitor korporasi tidak punya legitimasi politik.
- Moat 3 - Zero Waste Certified: Standar zero waste nasional yang MERLIN tulis sendiri. Kompetitor belum bersertifikat.
- Moat 4 - Blue Carbon Monetized: 12 juta ton CO2/tahun = USD 600 juta revenue carbon credit. Kompetitor baru pilot.

- Moat 5 - Riset Ekosistem: 50+ universitas, BRIN, CEDUS-UNHAS, 1000+ peneliti. Kompetitor tidak punya skala riset ini.
- Moat 6 - 100+ Produk Hilir: Diversifikasi produk dari food, pharma, kosmetik, bioplastic, bioenergy, biomaterial. Kompetitor fokus 5-10 produk.
- Moat 7 - Harga Jaminan Petani: Rp18.000/kg vs pasar Rp6.000/kg. Loyalti petani 100% ke MERLIN. Kompetitor sulit merebut supply.
- Moat 8 - Traceability Blockchain: Setiap produk dapat ditelusuri dari petani hingga shelf. Kompetitor masih manual.
- Defensibility Strategy: Paten 1000+ produk hilir, trademark MERLIN global, eksekusi cepat first-mover, lock-in offtake 10 tahun.

Owner: CSO + Legal Counsel MERLIN

PGA-16: Scenario Planning & Strategic Options

Tujuan: Perencanaan skenario strategis MERLIN untuk berbagai kondisi makro.

Elemen Kunci:

- Skenario 1 - Best Case (Maritime Renaissance): UU Rumput Laut disahkan 2026, investasi Rp800 triliun terkumpul, 50 EBIP 2030, ekspor Rp600 triliun. Probabilitas 25%.
- Skenario 2 - Base Case (Steady Growth): UU disahkan 2027, investasi Rp680 triliun, 40 EBIP 2030, ekspor Rp500 triliun. Probabilitas 55%.
- Skenario 3 - Worst Case (Setback): UU tertunda, investasi Rp400 triliun, 25 EBIP 2030, ekspor Rp300 triliun. Probabilitas 15%.
- Skenario 4 - Black Swan (Climate Crisis): Ocean acidification, ice-ice disease, El Nino ekstrem. Mitigasi: bibit tahan stress, diversifikasi lokasi, asuransi iklim. Probabilitas 5%.
- Decision Trees: Untuk setiap skenario, keputusan strategis berbeda (capex, ekspansi, hiring, M&A).
- Contingency Plans: Cadangan kas 12 bulan operasional, 3 sumber pendapatan alternatif, 5 negara tujuan ekspor cadangan.
- Early Warning Indicators: Harga komoditas, iklim, regulasi, kompetitor, teknologi. Dashboard real-time monitoring.
- Adaptive Strategy Framework: Quarterly strategy review, dynamic resource allocation, agile organizational structure.

Owner: CSO + Risk Committee MERLIN

◆ Akhir Domain 2 — 8 Dokumen PGA ◆

DOMAIN 3 | GOVERNANCE & LEGAL

Tata Kelola MERLIN — Bagaimana Keputusan Dibuat dan Akuntabilitas Dijaga

Range: PGA-17 s/d PGA-24 | Total: 8 Dokumen

PGA-17: Governance Charter MERLIN

Tujuan: *Piagam tata kelola MERLIN yang mengatur struktur, peran, dan proses pengambilan keputusan.*

Elemen Kunci:

- **Filosofi Governance:** Tri Hita Karana Laut (harmoni Tuhan-manusia-laut), transparency, accountability, responsibility, independency, fairness (TAIRIF).
- **Prinsip Governance:** (1) Dewan Pembina sebagai penjaga visi; (2) Dewan Pengawas sebagai pengawas independen; (3) Dewan Pakar sebagai penasihat teknis; (4) Eksekutif sebagai pelaksana operasional; (5) Member sebagai pemilik sejati.
- **Struktur Governance:** Dewan Pembina (Presiden RI + 5 menteri) → Dewan Pengawas (9 independen) → Dewan Pakar (46 lintas bidang) → Ketua Umum + CEO → 10 Ketua Asosiasi → Member.
- **Board Composition:** Dewan Pembina 6 orang, Dewan Pengawas 9 independen, Dewan Pakar 46 ahli, Eksekutif 1 CEO + 10 C-level.
- **Committee Structure:** Audit Committee, Risk Committee, Nomination & Remuneration Committee, ESG Committee, Ethics Committee, Strategy Committee.
- **Accountability Framework:** Setiap level memiliki KPI, laporan tahunan, audit eksternal, publikasi tahunan.
- **Governance Processes:** Rapat tahunan member, rapat quarterly board, rapat monthly eksekutif, emergency meeting protocol.

Owner: Sekretariat Dewan MERLIN

PGA-18: Organizational Structure MERLIN

Tujuan: *Struktur organisasi lengkap MERLIN dari level tertinggi hingga petani.*

Elemen Kunci:

- **Org Chart Level 1:** Dewan Pembina → Dewan Pengawas → Dewan Pakar → Ketua Umum MERLIN.
- **Org Chart Level 2:** CEO MERLIN Holding → COO, CFO, CTO, CMO, CSO, CHRO, Chief Legal, Chief Ecosystem.
- **Org Chart Level 3:** 10 Asosiasi (APRLN, AKNBB, AIPRL, ABBR, ABBION, APPOTL, APBB, AKBIN, ARTK, APERL) masing-masing dengan Ketua + Sekretaris + Bendahara.

- Org Chart Level 4: 40 Eco Blue Industrial Park, masing-masing dengan General Manager + 5 manager (Produksi, Quality, Logistics, HR, Finance).
- Org Chart Level 5: Koperasi MERLIN per kabupaten (350 koperasi), masing-masing dengan Ketua + 4 pengurus.
- Org Chart Level 6: Kelompok Tani MERLIN (10.000 kelompok), masing-masing dengan Ketua + 20-50 anggota.
- Span of Control: Setiap manajer 7-10 bawahan, ketua kelompok tani 20-50 petani, ketua koperasi 30 kelompok tani.
- Geographic Structure: Pusat (Jakarta), 4 Regional (Sumatera, Jawa, Kalimantan-Indonesia Tengah, Indonesia Timur), 34 Provinsi, 350 Kabupaten.
- Matrix Relationships: Asosiasi × Regional × Fungsi. Contoh: APRLN Sumatera Utara melapor ke Ketua APRLN Pusat + Regional Head Sumatera.

Owner: CHRO MERLIN

PGA-19: Role & Authority Matrix (RACI)

Tujuan: Matriks RACI untuk seluruh proses kunci MERLIN.

Elemen Kunci:

- Decision Rights Framework: Strategic (Dewan Pembina), Tactical (CEO + Dewan Pakar), Operational (Asosiasi + EBIP Manager), Routine (Koperasi + Kelompok Tani).
- RACI Matrix Key Processes: (1) Investasi >Rp1 triliun: R=CEO, A=Dewan Pembina, C=Dewan Pakar+Pengawas, I=Member; (2) Harga jaminan petani: R=APRLN, A=Dewan Pembina, C=AKNBB+APERL, I=Member; (3) Standar kualitas: R=AIPRL, A=Dewan Pengawas, C=ARTK, I=Semua; (4) Ekspor >USD 100 juta: R=APERL, A=CEO, C=AIPRL+Legal, I=Dewan.
- Authority Levels: <Rp1 miliar (EBIP Manager), Rp1 miliar-Rp10 miliar (Ketua Asosiasi), Rp10 miliar-Rp100 miliar (CEO), Rp100 miliar-Rp1 triliun (Dewan Pengawas), >Rp1 triliun (Dewan Pembina).
- Escalation Matrix: Routine → Operational → Tactical → Strategic → Emergency Board. SLA 24-72 jam per level.
- Delegation Framework: Setiap delegasi tertulis, batas jelas, reporting wajib, audit tahunan.
- Accountability Matrix: Setiap orang punya 1 supervisor langsung, 3-5 KPI, laporan mingguan, review quarterly.

Owner: Chief Legal + CHRO MERLIN

PGA-20: Decision-Making Protocol MERLIN

Tujuan: Protokol pengambilan keputusan terstruktur dengan kualitas terjamin.

Elemen Kunci:

- Decision Types: Strategic (5 tahun+), Tactical (1-5 tahun), Operational (1-12 bulan), Routine (harian/mingguan), Crisis (emergency).

- Decision Bodies: Dewan Pembina (strategic), Executive Committee (tactical), Asosiasi Board (operational), EBIP Manager (routine), Crisis Team (crisis).
- Decision Criteria: Financial (ROI, NPV, IRR), Strategic (alignment visi), Risk ($\leq$ toleransi), ESG (compliance), Stakeholder (acceptance).
- Decision Process: Problem definition → Data gathering → Option generation → Analysis → Decision → Implementation → Review.
- Documentation Requirements: Decision memo, supporting data, rationale, dissenting opinions, implementation plan.
- Review & Appeal: Setiap keputusan dapat di-review dalam 90 hari. Appeal ke level atas dengan justifikasi.
- Decision Quality Assurance: Pre-mortem analysis, red team challenge, expert consultation, member feedback.

Owner: Chief of Staff MERLIN

PGA-21: Legal Compliance Pack MERLIN

Tujuan: Kerangka kepatuhan hukum MERLIN terhadap seluruh regulasi nasional dan internasional.

Elemen Kunci:

- Regulatory Landscape: UU Koperasi, UU Perusahaan, UU Perikanan, UU Pariwisata, UU Perlindungan Konsumen, UU Persaingan Usaha, UU Perlindungan Data Pribadi, UU Cipta Kerja, Perpres Blue Economy, UU Rumpun Laut (draft).
- Compliance Requirements: 87 regulasi nasional + 45 regulasi internasional (EU, US, Japan, China, Codex).
- Compliance Framework: Policy → Procedure → Training → Monitoring → Audit → Remediation.
- Monitoring & Reporting: Monthly compliance report, quarterly audit, annual external audit, regulatory liaison.
- Training & Awareness: Annual compliance training, role-specific modules, certification, refresh quarterly.
- Audit & Assessment: Internal audit semester, external audit tahunan, regulatory inspection, ISO 37001 (anti-bribery), ISO 19650 (governance).
- Remediation Process: Root cause analysis, corrective action, preventive action, verification, closure.

Owner: Chief Legal Officer MERLIN

PGA-22: Constitutional Documents (AD/ART) MERLIN

Tujuan: Anggaran Dasar dan Rumah Tangga MERLIN sebagai konstitusi organisasi.

Elemen Kunci:

- Anggaran Dasar (AD): 18 bab meliputi nama, kedudukan, tujuan, keanggotaan, hak-kewajiban, struktur, rapat, keuangan, pembubaran.

- Anggaran Rumah Tangga (ART): 24 bab operasional meliputi tata cara rapat, pengelolaan dana, sanksi, pemilihan, musyawarah.
- Bylaws: Operational bylaws untuk 10 asosiasi, 40 EBIP, 350 koperasi, 10.000 kelompok tani.
- Member Agreements: Perjanjian keanggotaan dengan hak-kewajiban, harga jaminan, asuransi, pendidikan, exit clause.
- Constitutional Amendments: Hanya melalui Musyawarah Nasional dengan quorum 2/3 member, persetujuan 3/4 hadir.
- Interpretation Guidelines: Dewan Pengawas sebagai interpreter, preseden tertulis, kasus hukum.

Owner: Dewan Pengawas + Notaris MERLIN

PGA-23: Contracting Framework MERLIN

Tujuan: Kerangka kontrak standar untuk seluruh transaksi MERLIN.

Elemen Kunci:

- Contract Types: Offtake agreement, supply agreement, joint venture, licensing, distribution, employment, lease, construction, R&D.
- Contract Templates: 50+ template standar untuk setiap jenis kontrak, multilingual (ID, EN, ZH, JA).
- Negotiation Guidelines: BATNA, reservation price, ZOPA, walk-away points, escalation.
- Approval Process: Legal review → Business review → Risk review → Signatory. Threshold-based approval.
- Risk Allocation: Liability caps, indemnification, warranty, force majeure, dispute resolution.
- Performance Monitoring: KPI per kontrak, quarterly review, SLA tracking, penalty/bonus.
- Dispute Resolution: Mediation → Arbitration (BANI/ICC) → Litigation. Forum selection, governing law.

Owner: Legal Department MERLIN

PGA-24: Policy Register & Legal Tracker MERLIN

Tujuan: Daftar kebijakan MERLIN dan tracker perkara hukum.

Elemen Kunci:

- Policy Index: 200+ kebijakan internal terstruktur (HR, Finance, Operations, Quality, ESG, IT, Legal).
- Policy Development Process: Proposal → Drafting → Consultation → Approval → Communication → Implementation → Review.
- Policy Approval Workflow: Drafting (tim) → Review (Legal) → Approval (Eksekutif/Dewan) → Communication (semua member).
- Policy Communication: Intranet, email blast, training, town hall, signed acknowledgment.
- Policy Compliance Tracking: Dashboard real-time, audit semester, sanksi pelanggaran.

- Policy Review Schedule: Tahunan untuk operational, 3 tahunan untuk strategic, ad-hoc untuk regulatory change.
- Legal Case Tracker: Litigation, arbitration, regulatory inquiry, compliance investigation, IP dispute.

Owner: Compliance Officer MERLIN

◆ Akhir Domain 3 — 8 Dokumen PGA ◆

DOMAIN 4 | PRODUCT / SERVICE / SOLUTION

Arsitektur Produk & Layanan MERLIN — 100+ Produk Hilir

Range: PGA-25 s/d PGA-32 | Total: 8 Dokumen

PGA-25: Product/Service Architecture MERLIN

Tujuan: *Arsitektur produk 100+ turunan rumput laut dalam 7 kategori.*

Elemen Kunci:

- Kategori 1 - Food Ingredient (25 produk): Karragenan (refined, semi-refined, PES), Agar (food, bacteriological), Alginat (food, propylene glycol), jelly powder, pudding, ice cream stabilizer.
- Kategori 2 - Pharmaceutical (15 produk): Capsule shell, tablet binder, wound dressing, drug delivery, antiviral, antitumor, vaccine adjuvant, dental impression.
- Kategori 3 - Cosmeceutical (15 produk): Face mask, serum, cream, shampoo, soap, scrub, sunblock, anti-aging, whitening, acne.
- Kategori 4 - Bioplastic & Biomaterial (15 produk): Film packaging, bag, cutlery, textile fiber, paper coating, 3D printing filament, biocomposite.
- Kategori 5 - Bioenergy (10 produk): Bioethanol E30, biogas, biochar, bio-oil, green hydrogen feedstock.
- Kategori 6 - Agriculture & Feed (10 produk): Pupuk cair, biostimulant, pakan udang, pakan ikan, pakan abalone, pakan ternak.
- Kategori 7 - Nutraceutical & Supplement (10 produk): Fucoidan, laminarin, fucoxanthin, omega-3 DHA/EPA, iodine supplement, fiber supplement, weight management.

Owner: *Chief Product Officer MERLIN*

PGA-26: Product Requirement Document (PRD)

Tujuan: *Dokumen kebutuhan produk untuk 100+ produk hilir MERLIN.*

Elemen Kunci:

- PRD per Produk: Spesifikasi teknis, kualitas (USP/EP/FCC), volume target, harga target, target market, regulatory.
- PRD Karragenan Refined: Kappa (gel strength 800g/cm²), Iota (elastic), Lambda (non-gel). Volume 50.000 ton/tahun, harga USD 15/kg, market EU food.
- PRD Bioethanol E30: 99.5% purity, denatured, blend 30%. Volume 1 juta kiloliter/tahun, harga Rp 12.000/liter, market domestic fuel.
- PRD Bioplastic Film: Tensile 25 MPa, elongation 400%, biodegradable 90%/180 hari. Volume 100.000 ton/tahun, harga USD 4/kg, market EU packaging.
- PRD Fucoidan: 85% purity, food grade. Volume 500 ton/tahun, harga USD 500/kg, market Japan pharma.

- PRD Blue Carbon Credit: 1 ton CO2 equivalent, Verra VCS certified. Volume 12 juta ton/tahun, harga USD 50/ton, market global compliance.

Owner: Product Manager per kategori

PGA-27: Product Strategy & Roadmap

Tujuan: Strategi pengembangan produk MERLIN 2026-2030.

Elemen Kunci:

- Fase 1 (2026): 25 produk core (karragenan, agar, alginat, bioethanol, bioplastic basic).
- Fase 2 (2027): 50 produk expanded (cosmetic, nutraceutical, pharmaceutical basic).
- Fase 3 (2028): 75 produk advanced (biomaterial, specialty pharma, blue carbon).
- Fase 4 (2029): 100 produk diversified (genomic medicine, advanced biomaterial).
- Fase 5 (2030): 120 produk complete (full spectrum seaweed-derived products).
- Roadmap Technology: 2026 enzymatic extraction, 2027 supercritical CO2, 2028 biorefinery, 2029 synthetic biology, 2030 marine biotech hub.
- Innovation Pipeline: 200+ ide produk, 50 dalam R&D, 15 dalam pilot, 10 launch per tahun.

Owner: Chief Product + R&D Officer

PGA-28: System Architecture / Service Blueprint

Tujuan: Blueprint sistem dan layanan MERLIN terintegrasi.

Elemen Kunci:

- Sistem Produksi: Budidaya (longline, raft, IMTA) → Harvest → Drying → Processing → Refining → Formulation → Packaging.
- Sistem Digital: MERLIN Platform (ERP, CRM, SCM, BI), IoT aquaculture monitoring, blockchain traceability, MRV carbon.
- Sistem Layanan: Member portal, e-learning, telemedicine, financial services, marketplace B2B/B2C.
- Sistem Logistik: Cold chain, smart warehouse, dry port, maritime logistics, export handling.
- Sistem Kualitas: Lab nasional, sertifikasi, audit, continuous improvement.
- Sistem ESG: MRV carbon, zero waste monitoring, social impact, biodiversity tracking.

Owner: CTO + COO MERLIN

PGA-29: Customer/Member/User Journey Map

Tujuan: Pemetaan journey member MERLIN dari onboarding hingga loyalty.

Elemen Kunci:

- Petani Journey: Awareness (sosialisasi) → Onboarding (daftar koperasi) → Training (akademi) → Budidaya (bibit, alat) → Harvest (collecting) → Payment (harga jaminan) → Welfare (asuransi, pendidikan anak) → Loyalty (member 5 tahun+).

- Offtaker Journey: Awareness (trade show) → Inquiry (RFI) → Sample → Quality test → Negotiation → Contract (5-10 tahun) → Delivery → Payment → Renewal.
- Investor Journey: Awareness (roadshow) → Due diligence → Investment → Monitoring (quarterly report) → Exit (10 tahun).
- Tourist Journey: Awareness (digital marketing) → Booking → Arrival → Experience (resort, edutour, museum) → Share → Return.
- Pain Points per Journey: Petani (bibit mahal, harga tidak stabil); Offtaker (quality inconsistent); Investor (data transparency); Tourist (accessibility).
- Solutions per Pain Point: Bibit subsidi MERLIN, harga jaminan Rp18.000/kg, lab kualitas, dashboard investor, transportasi wisata.

Owner: CX Officer MERLIN

PGA-30: Quality Standard & Acceptance Criteria

Tujuan: Standar kualitas MERLIN dan kriteria penerimaan produk.

Elemen Kunci:

- Standar Kualitas MERLIN: Lebih tinggi dari internasional. Karragenan MERLIN $\geq$ gel strength 850g/cm² (international 800g). Moisture $\leq$ 8% (international 12%).
- Acceptance Criteria: Raw material (rumput laut kering moisture 20%, clean 95%), In-process (pH 7-8, temperature controlled), Finished (spec per PRD).
- Certification Targets: ISO 9001, ISO 22000, ISO 14001, ISO 45001, HACCP, Halal MUI, Kosher, FDA registered, EU organic, ASC-MSC seaweed.
- Quality Control Process: Incoming QC, In-process QC, Final QC, Release QC, Stability test.
- Quality Assurance: CAPA system, change control, deviation management, complaint handling, recall protocol.
- Continuous Improvement: Six Sigma DMAIC, Kaizen, root cause analysis, SPC, customer feedback loop.

Owner: Chief Quality Officer MERLIN

PGA-31: Innovation Pipeline MERLIN

Tujuan: Pipeline inovasi produk dan teknologi MERLIN.

Elemen Kunci:

- Pipeline Stage: Idea (200+) → Concept (50) → Feasibility (25) → Pilot (15) → Launch (10/year) → Scale (5/year) → Mature (100+).
- Innovation Themes: (1) Marine biotech; (2) Blue carbon; (3) Circular economy; (4) Digital aquaculture; (5) Functional food; (6) Sustainable packaging; (7) Bioenergy.
- Innovation Sources: Internal R&D (60%), University partnership (20%), Open innovation (10%), Member ideas (5%), Acquisition (5%).

- Innovation Process: Stage-gate with go/no-go decision per stage. Financial, technical, market criteria.
- Innovation Budget: 5% revenue = Rp25-42 miliar/tahun pada 2030.
- Innovation KPI: 10 new products/year, 50% revenue from products <5 years, 1000+ patents, 50+ publications.

Owner: Chief Innovation Officer MERLIN

PGA-32: Change Request & Version Governance

Tujuan: Tata kelola perubahan produk dan versi.

Elemen Kunci:

- Change Request Process: Submit → Impact analysis → Approval → Implementation → Verification → Documentation.
- Change Types: Minor (labeling), Moderate (formula tweak), Major (new product), Critical (regulatory).
- Version Control: Semantic versioning (major.minor.patch), changelog, backward compatibility.
- Document Control: Master document, revision history, approval matrix, distribution list.
- Change Communication: Notification to all stakeholders, training update, customer notification if needed.
- Audit Trail: All changes logged, who-what-when-why, retention 10 years.

Owner: Quality + IT MERLIN

◆ Akhir Domain 4 — 8 Dokumen PGA ◆

DOMAIN 5 | OPERATIONS & DELIVERY

Operasi & Pengiriman MERLIN — Eksekusi dari Hulu ke Hilir

Range: PGA-33 s/d PGA-40 | Total: 8 Dokumen

PGA-33: Master Operational SOP MERLIN

Tujuan: *SOP operasional master untuk seluruh aktivitas MERLIN.*

Elemen Kunci:

- SOP Budidaya: Site selection → Seedling → Planting → Maintenance → Harvest → Post-harvest. 12 langkah detail.
- SOP Processing: Receiving → Washing → Drying → Grinding → Extraction → Refining → Drying → Packaging. 18 langkah.
- SOP Quality Control: Sampling → Testing → Release → Documentation. 9 langkah.
- SOP Logistics: Order → Picking → Packing → Shipping → Tracking → Delivery. 10 langkah.
- SOP Export: Order → Documentation → Customs → Shipping → Destination customs → Delivery. 12 langkah.
- SOP Blue Tourism: Booking → Arrival → Check-in → Activity → Check-out → Feedback. 8 langkah.
- SOP Carbon Verification: Baseline → Monitoring → Reporting → Verification → Issuance → Trading. 10 langkah.

Owner: COO MERLIN

PGA-34: Process Map End-to-End

Tujuan: *Pemetaan proses end-to-end dari budidaya hingga ekspor.*

Elemen Kunci:

- End-to-End Value Chain: Bibit (ARTK) → Budidaya (APRLN) → Harvest → Collecting (AKNBB) → Processing (AIPRL/ABBR/ABBION/APPOTL) → Quality → Packaging → Export (APERL) / Tourism (APBB) / Carbon (AKBIN).
- Process Time: Bibit 30 hari → Budidaya 45-60 hari → Harvest 1 hari → Collecting 1-3 hari → Processing 7-14 hari → Quality 2-3 hari → Export 7-14 hari. Total 100-120 hari.
- Process Touchpoints: 7 asosiasi, 40 EBIP, 350 koperasi, 10.000 kelompok tani.
- Process Bottlenecks: Drying (weather dependent), Logistics (infrastruktur), Regulatory (permit).
- Process Optimization: Automation, IoT, AI scheduling, vertical integration.
- Process KPI: Cycle time, throughput, quality yield, cost per unit, customer satisfaction.

Owner: COO + Process Excellence MERLIN

PGA-35: Service Delivery Manual

Tujuan: Manual pengiriman layanan MERLIN kepada member dan customer.

Elemen Kunci:

- Service Catalog: 50+ layanan (budidaya, processing, logistics, training, financing, healthcare, certification, carbon trading).
- Service Level Agreement (SLA): Budidaya consulting 24 jam response, processing 7-14 hari, export 7-14 hari, training 30 hari schedule, healthcare 1 jam response.
- Service Delivery Channels: In-person (kantor cabang), Online (portal, app), Phone (call center 24/7), Field (mobil layanan).
- Service Standards: Response time, quality, professionalism, language (ID + local + EN).
- Service Recovery: Complaint handling 24 jam, resolution 7 hari, compensation policy.
- Service Innovation: Digital self-service, AI chatbot, predictive service, personalization.

Owner: Chief Service Officer MERLIN

PGA-36: Procurement & Vendor Playbook

Tujuan: Playbook pengadaan dan manajemen vendor MERLIN.

Elemen Kunci:

- Procurement Categories: Raw material, equipment, services, IT, logistics, professional services.
- Vendor Selection: Registration → Qualification → Evaluation → Approval → Contract → Performance review.
- Vendor Tiering: Strategic (5+ years contract), Preferred (1-3 years), Transactional (per project).
- Procurement Process: Need identification → Specification → Sourcing → RFQ → Evaluation → Negotiation → PO → Delivery → Payment.
- Procurement Ethics: No bribery, conflict of interest declaration, fair competition, transparency.
- Vendor KPI: On-time delivery 95%, quality acceptance 98%, cost competitive, ESG compliance.

Owner: Chief Procurement Officer MERLIN

PGA-37: Partnership Playbook

Tujuan: Playbook manajemen kemitraan strategis MERLIN.

Elemen Kunci:

- Partner Types: Government, university, corporate, financial, international organization, NGO.
- Partner Lifecycle: Identification → Outreach → Due diligence → MoU → Collaboration → Review → Renewal/Exit.
- Partner Tiering: Platinum (strategic 10+ years), Gold (5-10 years), Silver (1-5 years), Bronze (project).
- Partnership Models: Joint venture, R&D collaboration, offtake agreement, distribution, licensing, co-branding.
- Partner Governance: Steering committee quarterly, KPI review, conflict resolution, exit clause.
- Partner Value Sharing: Revenue share, equity, royalty, milestone payment, IP sharing.

Owner: Chief Ecosystem Officer MERLIN

PGA-38: Incident & Escalation SOP

Tujuan: SOP penanganan insiden dan eskalasi MERLIN.

Elemen Kunci:

- Incident Classification: P1 (critical, business stop), P2 (major, significant impact), P3 (minor, limited impact), P4 (cosmetic).
- Incident Response: Detection → Logging → Classification → Assignment → Investigation → Resolution → Closure → Post-mortem.
- Escalation Matrix: P1 immediate C-level, P2 4 jam eksekutif, P3 24 jam manager, P4 routine.
- Crisis Team: CEO, COO, CTO, Legal, Comms, HR. War room activation for P1.
- Communication Protocol: Internal (email, SMS, dashboard), External (press release, social media, regulator notification).
- Post-Incident: Root cause analysis (5 Why, fishbone), corrective action, preventive action, lessons learned.

Owner: Chief Risk Officer MERLIN

PGA-39: Business Continuity & Disaster Recovery Plan

Tujuan: Rencana kontinuitas bisnis dan pemulihan bencana MERLIN.

Elemen Kunci:

- BCP Framework: Prevention → Preparedness → Response → Recovery → Restoration.
- Risk Scenarios: Natural disaster (tsunami, earthquake, El Nino), pandemic, cyber attack, supply chain disruption, regulatory change, political instability.
- Critical Business Functions: Production, logistics, payment, communication, healthcare member.
- Recovery Time Objective (RTO): Production 7 hari, Payment 1 hari, Communication 4 jam, Healthcare 1 jam.
- Recovery Point Objective (RPO): Data backup hourly, redundant systems, geographic diversity.
- DR Site: Primary Jakarta, Secondary Surabaya, Tertiary Bali. Cloud backup multi-region.
- Testing: Annual full DR test, semi-annual tabletop, quarterly component test.

Owner: Chief Risk + IT MERLIN

PGA-40: Field/Branch/Deployment SOP

Tujuan: SOP operasional lapangan/cabang MERLIN.

Elemen Kunci:

- Branch Network: 350 kantor cabang kabupaten, 34 kantor provinsi, 4 kantor regional, 1 kantor pusat.
- Branch Setup: Site selection → Lease/Build → Staffing → Equipment → IT setup → Soft opening → Grand opening.

- Daily Operations: Opening → Service delivery → Reporting → Closing. Standard checklist.
- Branch Staffing: Branch manager, 4-10 staff (service, admin, finance, field officer).
- Branch KPI: Member growth, service satisfaction, financial performance, ESG compliance.
- Field Operations: Mobile service units, scheduled visit to kelompok tani, emergency response.

Owner: Head of Branch Network MERLIN

◆ Akhir Domain 5 — 8 Dokumen PGA ◆

DOMAIN 6 | FINANCE, RISK & CONTROL

Keuangan, Risiko & Kontrol MERLIN — Rp680 Triliun Investasi

Range: PGA-41 s/d PGA-48 | Total: 8 Dokumen

PGA-41: Financial Model / Projection MERLIN

Tujuan: Model keuangan MERLIN 2026-2035 dengan proyeksi lengkap.

Elemen Kunci:

- Revenue Projection: 2026 Rp5T → 2027 Rp25T → 2028 Rp80T → 2029 Rp200T → 2030 Rp500T → 2035 Rp1.200T.
- Cost Structure: COGS 55%, OpEx 25%, Depreciation 8%, Interest 4%, Tax 8%.
- EBITDA: 2030 Rp125T (margin 25%), 2035 Rp300T (margin 25%).
- Net Income: 2030 Rp50T (margin 10%), 2035 Rp120T (margin 10%).
- Balance Sheet: Asset 2030 Rp850T, Liability Rp400T, Equity Rp450T.
- Cash Flow: Operating +Rp80T, Investing -Rp100T, Financing +Rp50T, Net +Rp30T (2030).
- Key Assumptions: Produksi 25 juta ton, harga rata-rata Rp20.000/kg produk, 100+ produk mix, ekspor 70%.

Owner: CFO MERLIN

PGA-42: Unit Economics MERLIN

Tujuan: Ekonomi unit per produk dan per EBIP.

Elemen Kunci:

- Per EBIP: Revenue Rp2,5T/tahun, COGS Rp1,6T, Gross Profit Rp900M, OpEx Rp275M, EBITDA Rp625M (margin 25%).
- Per Ton Rumput Laut: Bahan baku Rp18.000/kg, Processing Rp5.000/kg, Logistics Rp2.000/kg, Total cost Rp25.000/kg, Selling price Rp45.000/kg (avg), Margin Rp20.000/kg.
- Per Produk Premium (Karragenan): Selling USD 15/kg, Cost USD 9/kg, Margin USD 6/kg (40%).
- Per Ton Carbon Credit: Revenue USD 50, Cost MRV USD 8, Margin USD 42 (84%).
- Per Tourist (Blue Tourism): Revenue USD 500/visit, Cost USD 300, Margin USD 200 (40%).
- Per Member Petani: Revenue contribution Rp30 juta/tahun, Cost Rp20 juta (training, insurance, etc), Margin Rp10 juta/tahun.

Owner: CFO + Strategy MERLIN

PGA-43: Budgeting & Cash Control SOP

Tujuan: SOP budgeting dan kontrol kas MERLIN.

Elemen Kunci:

- Budgeting Cycle: Annual (Q4 prior year), Quarterly review, Monthly tracking.
- Budget Process: Top-down guidance → Bottom-up planning → Negotiation → Approval → Communication → Monitoring.
- Cash Management: 12-month cash forecast, 3-month rolling, daily cash position.
- Payment Approval: <Rp1M manager, Rp1M-100M director, Rp100M-1B CEO, >Rp1B Board.
- Bank Accounts: Centralized treasury, 4 banks (Mandiri, BRI, BCA, Standard Chartered), multi-currency.
- Internal Control: Segregation of duties, dual authorization, monthly reconciliation, audit semester.

Owner: Treasurer MERLIN

PGA-44: Revenue Model & Monetization Playbook

Tujuan: Model pendapatan dan strategi monetisasi MERLIN.

Elemen Kunci:

- Revenue Streams: (1) Product sales 60%; (2) Tourism 15%; (3) Carbon credit 10%; (4) Licensing 5%; (5) Service fee 4%; (6) SaaS 3%; (7) Certification 3%.
- Pricing Strategy: Premium 15-25% di atas market (brand value), volume discount 5-15%, long-term contract 5-10% discount.
- Monetization Levers: Product premiumization, vertical integration, ecosystem network effect, IP licensing, carbon monetization, tourism upsell.
- Revenue Recognition: Per IFRS 15 / PSAK 72, point in time vs over time, contract asset/liability.
- Working Capital Optimization: Receivable 45 hari, Inventory 60 hari, Payable 30 hari, Cash conversion 75 hari.
- Revenue Diversification: No single product >15% revenue, no single customer >10%, no single market >30%.

Owner: CRO (Chief Revenue Officer) MERLIN

PGA-45: Risk Register & Mitigation Plan

Tujuan: Daftar risiko MERLIN dan rencana mitigasinya.

Elemen Kunci:

- Risk Categories: Strategic, Operational, Financial, Compliance, ESG, Cyber, Geopolitical, Climate.
- Top 12 Risks: (1) Climate change impact on cultivation; (2) Regulatory uncertainty; (3) Currency fluctuation; (4) Supply chain disruption; (5) Cyber attack; (6) Pest/disease (ice-ice); (7) Competitor pricing; (8) Quality failure; (9) Talent shortage; (10) ESG non-compliance; (11) Political instability; (12) Technology obsolescence.
- Risk Assessment: Likelihood (1-5) × Impact (1-5) = Risk Score. Red ≥15, Yellow 8-14, Green ≤7.
- Mitigation per Risk: Climate (bibit tahan stress, diversifikasi lokasi, asuransi), Regulatory (lobby, compliance team), Currency (hedging), etc.

- Risk Owner: Setiap risiko punya owner, mitigation plan, deadline, status reporting.
- Risk Monitoring: Quarterly risk review, early warning indicators, escalation protocol.

Owner: Chief Risk Officer MERLIN

PGA-46: Audit & Internal Control Framework

Tujuan: Kerangka audit dan kontrol internal MERLIN.

Elemen Kunci:

- Internal Audit: Charter, annual plan, risk-based audit, field work, report, follow-up.
- External Audit: Big-4 audit firm, annual financial audit, ESG audit, special purpose audit.
- Control Framework: COSO Internal Control, COSO ERM, ISO 31000, COBIT for IT.
- Key Controls: Segregation of duties, authorization, reconciliation, physical safeguard, IT general control.
- Control Testing: Quarterly self-assessment, semi-annual internal audit, annual external audit.
- Audit Committee: 3 independent members, quarterly meeting, direct report to Board.

Owner: Internal Audit + Audit Committee

PGA-47: Investment / Capital / Funding Memorandum

Tujuan: Memorandum investasi MERLIN Rp680 triliun.

Elemen Kunci:

- Capital Structure: Equity 40% (Rp272T), Debt 45% (Rp306T), Grant/Subsidy 15% (Rp102T).
- Equity Sources: Government 25%, Strategic investor 35%, Public IPO 20%, Member cooperative 15%, CSR 5%.
- Debt Sources: Bank loan (Mandiri, BRI) 40%, Bond (sukuk) 30%, Multilateral (IFC, ADB, World Bank) 20%, Export credit 10%.
- Use of Proceeds: EBIP infrastructure 35%, Pabrik 27%, Blue Tourism 22%, Logistics 7%, R&D 5%, Brand 4%.
- Return Profile: IRR 18%, payback 7 tahun, NPV Rp450T (discount 12%), EBITDA margin 25%.
- ESG Compliance: IFC Performance Standards, Equator Principles, EU Taxonomy, SDG alignment.
- Exit Options: IPO MERLIN Holding 2032, strategic sale, secondary buyout, recapitalization.

Owner: CFO + Investment Committee

PGA-48: Treasury / Asset / Reserve Policy

Tujuan: Kebijakan treasury, aset, dan cadangan MERLIN.

Elemen Kunci:

- Treasury Policy: Cash management, investment of surplus, foreign exchange, interest rate.
- Asset Management: Fixed asset register, depreciation policy, asset utilization, disposal.

- Reserve Policy: Operating reserve 6 months, Capex reserve 12 months, Insurance reserve, Sovereign fund 5% revenue.
- Investment Policy: Conservative (cash, bond), Moderate (equity 30%), Strategic (EBIP expansion).
- FX Policy: Natural hedge, forward contract, option, 70% hedged for 12-month exposure.
- Insurance: Property, business interruption, liability, marine cargo, cyber, climate, crop.

Owner: *Treasurer + Investment Committee*

◆ *Akhir Domain 6 — 8 Dokumen PGA* ◆

DOMAIN 7 | PEOPLE, CULTURE & LEADERSHIP

SDM, Budaya & Kepemimpinan MERLIN — 5 Juta Tenaga Kerja

Range: PGA-49 s/d PGA-56 | Total: 8 Dokumen

PGA-49: Talent & Recruitment Framework MERLIN

Tujuan: Kerangka rekrutmen dan talent management untuk 5 juta tenaga kerja MERLIN.

Elemen Kunci:

- Workforce Planning 2030: 5 juta tenaga kerja - 1,5 juta petani, 1 juta industri hilir, 500rb pariwisata, 500rb logistik, 500rb support, 1 juta koperasi & retail.
- Recruitment Channels: Internal (member), External (job portal, campus), Referral, Specialized (executive search).
- Selection Process: Application → Screening → Test (aptitude, technical, cultural) → Interview (3 round) → Background check → Offer → Onboarding.
- Talent Pool: Database 1 juta kandidat, succession planning 3 deep per critical role.
- Diversity Target: 50% women, 30% local community, 5% disability, 100% Indonesian (except specialized expat <2%).
- Employer Branding: MERLIN as purpose-driven employer, competitive package, career growth, work-life balance.

Owner: CHRO MERLIN

PGA-50: Leadership Model MERLIN

Tujuan: Model kepemimpinan MERLIN untuk seluruh level.

Elemen Kunci:

- Leadership Philosophy: Servant-leader (melayani member), Sage (ilmuwan), Sovereign (berdaulat), Steward (amanah).
- Leadership Competencies: 8 kompetensi - Strategic thinking, People development, Decision making, Innovation, Integrity, Resilience, Cultural sensitivity, ESG mindset.
- Leadership Pipeline: Individual → Supervisor → Manager → Director → C-level → Board. 6 level dengan clear progression criteria.
- Leadership Development: Akademi MERLIN, executive coaching, mentor program, rotation, international assignment.
- Leadership Assessment: 360-degree feedback annually, potential matrix (9-box), development plan.
- Succession: 3 deep per critical role, emergency succession, planned succession 2-5 years ahead.

Owner: CHRO + CEO MERLIN

PGA-51: HR Operating Manual

Tujuan: *Manual operasional HR MERLIN.*

Elemen Kunci:

- Employment Lifecycle: Recruitment → Onboarding → Development → Performance → Compensation → Engagement → Retention → Exit.
- Compensation & Benefits: Base salary (market median P75), bonus (performance 1-4 months), benefits (health, pension, insurance, education), member benefits (16 hak member).
- Performance Management: Annual goal setting, mid-year review, year-end review, calibration, development plan.
- Learning & Development: 40 jam/tahun training, internal Akademi, external certification, tuition assistance.
- Employee Relations: Grievance procedure, conflict resolution, labor union engagement, collective agreement.
- HRIS: SAP SuccessFactors or Workday, payroll, self-service, analytics, mobile app.

Owner: *CHRO + HR Operations*

PGA-52: Performance Management System

Tujuan: *Sistem manajemen kinerja MERLIN.*

Elemen Kunci:

- Performance Framework: OKR (Objectives & Key Results) + KPI + Competency + Values.
- Goal Setting: Cascade from corporate → function → team → individual. Quarterly check-in, annual review.
- KPI Examples: Production volume, quality yield, cost efficiency, customer satisfaction, ESG score, innovation.
- Performance Rating: 5-point scale (1 unsatisfactory to 5 exceptional), forced distribution (10-20-40-20-10).
- Reward: Performance bonus 1-6 months, merit increase 5-15%, promotion, equity.
- Underperformer Management: PIP (Performance Improvement Plan) 90 hari, coaching, reassignment, exit.

Owner: *CHRO + Managers MERLIN*

PGA-53: Training & Certification Framework

Tujuan: *Kerangka pelatihan dan sertifikasi MERLIN.*

Elemen Kunci:

- Akademi MERLIN: 5 kampus utama (Jakarta, Surabaya, Makassar, Bali, Sorong), 40 training center per EBIP, e-learning platform.
- Curriculum: Budidaya (3 bulan), Processing (6 bulan), Quality (3 bulan), Leadership (12 bulan), Marine biotech (24 bulan MSc).

- Certification: 50+ sertifikat (Budidaya, Quality, Safety, HACCP, ISO, Leadership, Carbon MRV, Marine biotech).
- Trainer: 1000+ internal trainer, 500+ eksternal expert, partnership dengan 50+ universitas.
- Lifelong Learning: 40 jam/tahun mandatory, continuous certification renewal 3 tahun.
- Member Training: 1,5 juta petani trained 2026-2030, free of charge, with certification.

Owner: Chief Learning Officer MERLIN

PGA-54: Culture Code MERLIN

Tujuan: Kode budaya MERLIN yang mengikat seluruh ekosistem.

Elemen Kunci:

- Culture Pillars: Bahari (maritime spirit), Bersatu (unity), Berkelanjutan (sustainable), Berdaulat (sovereign), Berilmu (knowledge).
- Cultural Norms: Gotong royong, musyawarah-mufakat, hormat senior, kemitraan setara, akuntabilitas.
- Rituals: Upacara member baru, apel pagi, rapat mingguan, festival tahunan, penghargaan tahunan.
- Symbols: Logo MERLIN, seragam, lagu, hymne, bendera, badge per pencapaian.
- Language: Bahasa Indonesia resmi, bahasa daerah dihormati, English untuk internasional.
- Cultural Ambassador: Setiap kelompok tani, koperasi, EBIP punya 1 culture ambassador.

Owner: Chief Culture Officer MERLIN

PGA-55: Ethics & Conduct Code

Tujuan: Kode etik dan perilaku MERLIN.

Elemen Kunci:

- Ethical Principles: Integrity, honesty, fairness, respect, responsibility, transparency, sustainability.
- Anti-Bribery: Zero tolerance, ISO 37001, gift policy (max Rp500rb), conflict of interest declaration.
- Anti-Corruption: No facilitation payment, whistleblower protection, due diligence third party.
- Conflict of Interest: Declaration annually, recusal on conflict, second job approval.
- Confidentiality: Trade secret, member data, financial info. NDA, exit non-disclosure.
- Whistleblower: Hotline 24/7, anonymous, protection, investigation, reward.
- Sanctions: Warning → Suspension → Termination → Legal action. Published case library.

Owner: Ethics Committee + Compliance

PGA-56: Succession & Continuity Plan

Tujuan: Rencana suksesi dan kontinuitas kepemimpinan MERLIN.

Elemen Kunci:

- Critical Positions: CEO, 10 Ketua Asosiasi, 40 EBIP GM, 350 Koperasi Ketua, 1000+ senior leader.
- Succession Strategy: 3 deep per critical role, ready-now / ready-1-year / ready-2-year.
- Talent Review: Annual 9-box matrix, calibration session, development plan.
- Emergency Succession: Sudden departure, death, incapacity. 24-hour succession protocol.
- Planned Succession: 2-5 years transition, mentoring, gradual handover.
- Continuity: Knowledge management, SOP documentation, cross-training, rotation.

Owner: CHRO + Board Nomination Committee

◆ Akhir Domain 7 — 8 Dokumen PGA ◆

DOMAIN 8 | BRAND, GROWTH & STAKEHOLDERS

Brand, Pertumbuhan & Pemangku Kepentingan MERLIN

Range: PGA-57 s/d PGA-64 | Total: 8 Dokumen

PGA-57: Brand & Communication Guideline MERLIN

Tujuan: *Pedoman brand dan komunikasi MERLIN.*

Elemen Kunci:

- Brand Identity: MERLIN (Masyarakat Ekosistem Rumput Laut Indonesia). Tagline: "Laut Berkemakmuran, Nusantara Berdaulat".
- Visual Identity: Logo (gelombang + rumput laut + bintang), color (deep ocean blue #0A4D8C, seaweed green #007A5A, gold #C9A227), typography (Arial Bold heading, Times New Roman body).
- Brand Voice: Formal-profesional (dokumen), inspiratif-naratif (publik), teknis-presisi (industri), hangat-empatis (petani).
- Brand Application: Stationery, signage, uniform, vehicle, packaging, digital, advertising, events.
- Sub-Brands: 10 asosiasi (APRLN, AKNBB, AIPRL, ABBR, ABBION, APPOTL, APBB, AKBIN, ARTK, APERL), 40 EBIP (MERLIN Park 1-40), Blue Tourism (MERLIN Beach).
- Brand Guidelines: 100+ pages book, digital asset management, brand portal, training.

Owner: *Chief Brand Officer MERLIN*

PGA-58: Narrative & Messaging Matrix

Tujuan: *Matriks narasi dan messaging per audience.*

Elemen Kunci:

- Core Narrative: "MERLIN adalah gerakan kemandirian ekonomi maritim Indonesia, mengubah laut dari sumber kemiskinan menjadi sumber kemakmuran abadi."
- Audience Messaging: Petani (harga jaminan, sejahtera); Investor (IRR 18%, ESG); Government (5 juta kerja, devisa); Customer (premium sustainable); Media (maritime renaissance); Youth (karir bahari).
- Message House: Master narrative → 5 pillar messages → 20 proof points → 100 stories.
- Proof Points: 40 EBIP, 5 juta kerja, Rp500T ekspor, 12 juta ton CO2, 100+ produk.
- Stories: Petani Naik Kelas (Bapak Andi di Wakatobi), Eco Industrial Park (Model Bali), Blue Carbon (Pilot Derawan), dll.
- Message Testing: A/B testing, focus group, sentiment analysis, quarterly refresh.

Owner: *CMO + Comms MERLIN*

PGA-59: Community & Engagement Strategy

Tujuan: *Strategi komunitas dan engagement MERLIN.*

Elemen Kunci:

- Community Segments: Member (1,5 juta petani + 10 juta keluarga), Customer (B2B 5000+ corporate, B2C 1 juta+), Partner (5000+), Public (300 juta+ Indonesian), International (global seaweed community).
- Engagement Channels: Member portal, mobile app, town hall, festival, social media, newsletter, magazine, radio, TV.
- Engagement Programs: Member (rewards, leaderboard, scholarship, healthcare, pension); Customer (loyalty, education, sustainability report); Partner (joint event, co-marketing); Public (awareness, education, CSR).
- Community Building: 10.000 kelompok tani, 350 koperasi, 40 EBIP community center, 40 Blue Tourism community.
- Volunteer Program: 100.000 relawan MERLIN (teaching, mentoring, disaster response, environmental).
- Feedback Loop: Survey tahunan, suggestion box, complaint handling, co-creation.

Owner: *Chief Community Officer MERLIN*

PGA-60: Stakeholder Map MERLIN

Tujuan: *Pemetaan pemangku kepentingan MERLIN.*

Elemen Kunci:

- Stakeholder Categories: Internal (member, employee, board), Government (Presiden, 12 kementerian, DPR, 34 provinsi, 350 kabupaten), Industry (ARLI, partner, supplier, customer), Financial (investor, bank, rating), Society (NGO, media, community, academia), International (UN, FAO, World Bank, foreign gov).
- Power-Interest Matrix: High-High (manage closely - Presiden, investor besar); High-Low (keep satisfied - regulator); Low-High (keep informed - media); Low-Low (monitor - general public).
- Stakeholder Engagement Plan per group: Frequency, channel, message, owner, KPI.
- Stakeholder Issues: Petani (harga, kesejahteraan); Government (regulasi, devisa); Investor (return, ESG); Customer (quality, sustainability); NGO (environment, social).
- Stakeholder Communication: Annual report, quarterly briefing, monthly newsletter, real-time dashboard.
- Stakeholder Conflict Resolution: Mediation, escalation, joint problem solving.

Owner: *Chief Stakeholder Officer MERLIN*

PGA-61: Sales / Acquisition Playbook

Tujuan: *Playbook penjualan dan akuisisi customer MERLIN.*

Elemen Kunci:

- Sales Process: Lead → Qualify → Discover → Propose → Negotiate → Close → Onboard → Retain.
- Sales Team: 500+ sales rep (B2B key account, regional, inside sales), 50+ technical sales, 20+ export sales.
- Sales Tools: CRM (Salesforce), product catalog, technical datasheet, sample, presentation, ROI calculator.
- Sales Compensation: Base + commission (2-5% revenue) + bonus (target achievement) + retention bonus.
- Sales Pipeline: USD 5 billion pipeline 2030, 25% conversion, 12-month cycle average.
- Customer Acquisition Cost (CAC): USD 50k per B2B customer, payback 18 months, LTV/CAC ratio 8:1.
- Retention: 85% target, account management, quarterly business review, expansion selling.

Owner: Chief Sales Officer MERLIN

PGA-62: Public Affairs / Government Relations Framework

Tujuan: Kerangka hubungan pemerintah MERLIN.

Elemen Kunci:

- Government Stakeholders: Presiden, KKP, ESDM, Kemenparekraf, KLHK, Bappenas, Kemenkeu, Kemenperin, Kemendag, KemenkopUKM, Kemenkominfo, DPR (11 komisi).
- Policy Advocacy: UU Rumput Laut, Perpres Blue Economy, RPJMN, DAK, tax incentive, export policy.
- Lobbying Strategy: Direct meeting, written submission, hearing, coalition, media, public campaign.
- Government Relations Team: 20+ person, ex-government official, Jakarta-based, regional coverage.
- Compliance: Lobbyist registration, gift ban, transparency, anti-bribery.
- Success Metrics: UU Rumput Laut disahkan 2027, 10 policy improvement, Rp10T DAK, tax holiday 10 tahun.

Owner: Chief Public Affairs Officer MERLIN

PGA-63: Investor / Donor / Partner Deck

Tujuan: Deck presentasi untuk investor, donor, dan partner.

Elemen Kunci:

- Investor Deck (60 slides): Problem → Solution → Market → Business Model → Traction → Team → Financials → Ask → Use of Funds → Exit. USD 680 trillion ask.
- Donor Deck (40 slides): Mission → Impact → Track Record → Programs → Budget → Sustainability. For CSR, grant, blended finance.
- Partner Deck (30 slides): Vision → Ecosystem → Value Proposition → Partnership Models → Benefits → Process. For strategic alliance.

- Pitch Materials: One-pager, executive summary, full deck, financial model, due diligence pack, term sheet template.
- Roadshow: Annual investor roadshow (Singapore, Tokyo, London, NY, Dubai), quarterly partner forum, monthly donor briefing.
- Investor Relations: Quarterly earnings, annual report, ESG report, investor day, one-on-one meeting.

Owner: CFO + IR Officer MERLIN

PGA-64: Media & Crisis Communication SOP

Tujuan: SOP komunikasi media dan krisis MERLIN.

Elemen Kunci:

- Media Strategy: Proactive storytelling, thought leadership, ESG positioning, maritime renaissance narrative.
- Media Relations: 50+ key journalist (Bloomberg, Reuters, Nikkei, Tempo, Kompas, CNN Indonesia), press release monthly, press conference quarterly.
- Social Media: LinkedIn (B2B, 1 juta followers), Instagram (B2C, 5 juta), Twitter/X (public), TikTok (youth), YouTube (edutainment).
- Crisis Communication: Crisis team (CEO, Comms, Legal, HR, Operations), war room, holding statement 1 jam, full response 4 jam.
- Crisis Scenarios: Quality recall, environmental incident, leadership scandal, regulatory action, natural disaster.
- Reputation Management: Sentiment monitoring, brand tracker, response protocol, recovery plan.

Owner: Chief Communications Officer MERLIN

◆ Akhir Domain 8 — 8 Dokumen PGA ◆

DOMAIN 9 | DATA, INTELLIGENCE & LEGACY

Data, Intelijen & Warisan MERLIN — Aset Pengetahuan Abadi

Range: PGA-65 s/d PGA-72 | Total: 8 Dokumen

PGA-65: Data Governance & Privacy Policy MERLIN

Tujuan: Kebijakan tata kelola data dan privasi MERLIN.

Elemen Kunci:

- Data Classification: Public, Internal, Confidential, Restricted. 4 level dengan akses control berbeda.
- Data Lifecycle: Creation → Storage → Use → Archive → Destruction. Retention policy per data type.
- Data Quality: Accuracy, completeness, consistency, timeliness, validity. Master data management.
- Privacy Compliance: UU PDP Indonesia, GDPR (EU customer), CCPA (US), data minimization, consent management.
- Data Security: Encryption at rest/in transit, RBAC, MFA, DLP, audit log, incident response.
- Data Governance Body: Chief Data Officer, Data Council, Data Steward per domain.
- Data Monetization: Anonymized aggregate data for industry report, market intelligence, research partnership.

Owner: Chief Data Officer MERLIN

PGA-66: Security & Cyber Compliance Framework

Tujuan: Kerangka keamanan dan kepatuhan siber MERLIN.

Elemen Kunci:

- Security Framework: ISO 27001, NIST CSF, SOC 2 Type II, PCI DSS (if payment).
- Security Operations: SOC 24/7, SIEM, SOAR, threat intelligence, vulnerability management, penetration testing.
- Identity & Access: SSO, MFA, RBAC, PAM (privileged access), JIT access.
- Network Security: Firewall, IDS/IPS, WAF, DDoS protection, zero-trust architecture.
- Endpoint Security: EDR, MDM, patch management, asset inventory.
- Incident Response: Detection → Containment → Eradication → Recovery → Lessons learned. RTO 4 jam, RPO 1 jam.
- Compliance: ISO 27001 certification 2027, SOC 2 Type II 2028, annual recertification.

Owner: CISO MERLIN

PGA-67: KPI Dashboard & North Star Metrics

Tujuan: *Dashboard KPI dan North Star metrics MERLIN.*

Elemen Kunci:

- North Star Metric: Member Welfare Index (MWI) - composite of income, health, education, satisfaction. Target MWI 80/100 2030.
- Corporate KPIs: Revenue, EBITDA, member count, jobs created, export value, carbon sequestered, products launched, customer NPS.
- Functional KPIs: Production (ton/hektar), Quality (defect rate), Sales (conversion), HR (turnover), Finance (cash flow), ESG (carbon intensity).
- Dashboard Hierarchy: Board (10 KPI), Executive (50 KPI), Function (200 KPI), Operational (1000 KPI).
- Data Source: ERP, CRM, IoT, manual input. Real-time (operational), daily (tactical), monthly (strategic).
- Analytics: Descriptive (what happened), Diagnostic (why), Predictive (what next), Prescriptive (what to do).
- Review Cadence: Daily huddle, weekly review, monthly business review, quarterly board, annual strategy.

Owner: *Chief Analytics Officer MERLIN*

PGA-68: Knowledge Management & Documentation SOP

Tujuan: *SOP manajemen pengetahuan dan dokumentasi MERLIN.*

Elemen Kunci:

- Knowledge Taxonomy: Strategy, Operations, Technical, Member, Stakeholder, Lessons Learned, Best Practices.
- Knowledge Capture: After-action review, expert interview, document template, video, podcast.
- Knowledge Repository: SharePoint/Confluence, search engine, tagging, version control.
- Knowledge Sharing: Community of practice, brown bag, mentor, conference, publication.
- Knowledge Transfer: Onboarding, rotation, succession, exit interview.
- Lessons Learned: Project debrief, incident post-mortem, success story, failure analysis.
- Knowledge Metrics: Documentation coverage 95%, search success 90%, knowledge reuse 70%.

Owner: *Chief Knowledge Officer MERLIN*

PGA-69: Research, Benchmark & Intelligence Register

Tujuan: *Daftar riset, benchmark, dan intelijen MERLIN.*

Elemen Kunci:

- Research Agenda: 50 topik per tahun - budidaya (15), processing (10), product (10), ESG (5), market (5), tech (5).
- Benchmark Partners: Top seaweed companies (Cargill, CP Kelco, DuPont), top cooperative (Amul, FrieslandCampina), top ESG (Patagonia, Unilever).

- Competitive Intelligence: Quarterly competitor report (Filipina, China, Korea, Japan), pricing, product, M&A.
- Market Intelligence: Market size, growth, trend, customer insight, regulatory change - monthly report.
- Technology Intelligence: Emerging tech, patent landscape, startup ecosystem, academic research.
- Intelligence Sources: Database (Bloomberg, Crunchbase, Nielsen), field research, partner network, satellite imagery.
- Intelligence Distribution: Weekly brief, monthly report, quarterly forum, annual conference.

Owner: Chief Intelligence Officer MERLIN

PGA-70: ESG / Sustainability / Impact Framework

Tujuan: Kerangka ESG, keberlanjutan, dan dampak MERLIN.

Elemen Kunci:

- ESG Framework: Environmental (carbon, water, waste, biodiversity, energy), Social (member welfare, community, labor, health, education), Governance (board, ethics, transparency, risk, compliance).
- Environmental Targets: Carbon negative (12 juta ton CO2/tahun), zero waste, 100% renewable energy, water recycle 90%, biodiversity net positive.
- Social Targets: 5 juta jobs, member welfare index 80, gender equality 50%, zero poverty, education access 100%.
- Governance Targets: Board independence 50%, audit compliance 100%, ethics training 100%, transparency score 90+.
- Reporting: Annual ESG report (GRI, SASB, TCFD, TNFD), quarterly ESG dashboard, third-party assurance.
- Certification: B Corp 2028, ASC-MSC seaweed 2027, EU organic 2027, Verra VCS carbon 2027.
- Impact Measurement: SROI (Social Return on Investment) 5:1, theory of change, impact data per program.

Owner: Chief Sustainability Officer MERLIN

PGA-71: Legacy / 100-Year Covenant / Foundational Commitments

Tujuan: Warisan 100 tahun MERLIN dan komitmen fondasional.

Elemen Kunci:

- 100-Year Vision (2126): Indonesia sebagai maritime civilization superpower, MERLIN sebagai foundation abadi, ekosistem laut lestari, 100 juta rakyat sejahtera dari laut.
- Foundational Commitments: (1) Harga jaminan petani selamanya; (2) Zero waste selamanya; (3) Blue carbon positive selamanya; (4) Member welfare di atas profit; (5) Riset 5% revenue selamanya.

- Endowment Fund: MERLIN Legacy Fund Rp50 triliun (5% revenue/tahun), income untuk riset, member welfare, disaster relief.
- Heritage Assets: 40 EBIP sebagai heritage industrial, 40 Blue Tourism sebagai heritage cultural, Museum MERLIN.
- Legacy Stories: 1000+ stories per generasi (petani, worker, leader, partner), buku, film, museum.
- Legacy Governance: Legacy Council (10 elder), decennial review (2036, 2046...), 100-year audit.

Owner: Legacy Council MERLIN

PGA-72: Master Index & Document Control Register

Tujuan: Indeks master dan registrasi kontrol dokumen MERLIN.

Elemen Kunci:

- Document Hierarchy: 165 PGA → sub-documents → supporting documents. Total 5000+ dokumen.
- Document Control: Versioning (semantic), approval matrix, distribution, retention, destruction.
- Document Repository: SharePoint dengan metadata, search, version history, audit trail.
- Document Quality: Template, style guide, peer review, editorial calendar.
- Document Access: Role-based, watermarked, downloadable per role, time-limited link.
- Document Lifecycle: Draft → Review → Approve → Publish → Maintain → Archive → Retire.
- Document Audit: Annual document audit, compliance check, gap analysis, remediation.

Owner: Chief Document Controller MERLIN

◆ Akhir Domain 9 — 8 Dokumen PGA ◆

DOMAIN 10 | DIPLOMACY & INTERNATIONAL RELATIONS

Diplomasi & Hubungan Internasional MERLIN — Diplomasi Ekonomi Biru

Range: PGA-73 s/d PGA-80 | Total: 8 Dokumen

PGA-73: Bilateral Cooperation Framework

Tujuan: *Kerangka kerja sama bilateral MERLIN dengan negara lain.*

Elemen Kunci:

- Priority Bilateral: China (market + investment), Japan (technology + market), Korea (technology + market), EU (market + ESG), US (market + technology), Australia (research + carbon), UAE (investment + market), Saudi (investment + halal).
- Cooperation Models: Trade agreement, investment treaty, R&D collaboration, capacity building, carbon trade, scholarship exchange.
- Bilateral Roadmap: Per country 5-year roadmap with milestones, KPI, responsible party.
- Diplomatic Channels: Embassy, trade attaché, business delegation, state visit, ministerial meeting.
- Joint Commission: Annual bilateral joint commission co-chaired with counterpart.
- Success Metrics: 8 strategic bilateral 2030, USD 5 billion FDI, 20 R&D collaboration, 500 scholarship exchange.

Owner: *Chief Diplomatic Officer MERLIN*

PGA-74: International Arbitration Protocol

Tujuan: *Protokol arbitrase internasional MERLIN.*

Elemen Kunci:

- Arbitration Clause: Standard clause in all international contract. Choice of BANI, ICC, SIAC, HKIAC.
- Governing Law: Indonesian law for domestic, English/Singapore law for international, UNIDROIT principles.
- Arbitrator Selection: Independent arbitrator, expertise in maritime/trade, language capability.
- Procedure: Submission → Hearing → Award. Fast track 6 months, normal 18 months.
- Enforcement: New York Convention 1958, enforcement in 170+ countries.
- Cost Management: Arbitration cost budget, third-party funding, insurance.

Owner: *General Counsel MERLIN*

PGA-75: Diplomatic Immunity Manual

Tujuan: *Manual kekebalan diplomatik MERLIN (untuk perwakilan internasional).*

Elemen Kunci:

- Diplomatic Status: MERLIN representatives dengan status quasi-diplomatic untuk international negotiation.
- Immunity Scope: Official act immunity, travel facilitation, tax exemption (limited).
- Protocol: Vienna Convention on Diplomatic Relations 1961 as reference.
- Privilege Card: MERLIN diplomatic passport/card untuk perwakilan resmi.
- Limitation: No immunity for commercial activity, criminal act, traffic violation.
- Host Country Agreement: Bilateral agreement on MERLIN representative status.

Owner: Chief Diplomatic Officer + Legal

PGA-76: UN / ICA / ILO Engagement Strategy

Tujuan: Strategi keterlibatan MERLIN dengan UN, ICA, ILO.

Elemen Kunci:

- UN Engagement: UNDP (SDG partnership), FAO (fishery standard), UNEP (environment), UNIDO (industry), UN Global Compact member.
- ICA Engagement: International Co-operative Alliance membership, ICA Asia-Pacific, annual congress.
- ILO Engagement: Decent work agenda, labor standard, child labor elimination, gender equality.
- Other International: World Bank, ADB, OECD, WTO, ISO, Codex Alimentarius, MSC, ASC, RSB.
- Engagement Strategy: Annual forum, joint program, standard setting, advocacy, funding partnership.
- Success Metrics: 20 international membership, 10 joint program, 5 standard contribution, USD 500 juta funding.

Owner: International Affairs Director MERLIN

PGA-77: South-South Cooperation Doctrine

Tujuan: Doktrin kerja sama selatan-selatan MERLIN.

Elemen Kunci:

- South-South Partners: ASEAN (Philippines, Vietnam, Thailand, Malaysia), Pacific (Fiji, PNG, Solomon), Africa (Tanzania, Madagascar, Senegal, Morocco), Latin America (Chile, Peru, Brazil).
- Cooperation Themes: Technology transfer, capacity building, market access, climate adaptation, blue economy.
- Cooperation Models: Bilateral, triangular (with developed country partner), regional, multilateral.
- Indonesia Role: As leader in seaweed South-South, share MERLIN model, training, scholarship, technical assistance.
- Programs: 1000 scholarship for South countries, 100 expert deployment, 50 joint R&D, 10 demonstration EBIP.
- Funding: Indonesia South-South fund (SSCF), partner country fund, multilateral co-financing.

Owner: South-South Cooperation Director MERLIN

PGA-78: Diaspora Economic Integration

Tujuan: Integrasi ekonomi diaspora dengan MERLIN.

Elemen Kunci:

- Diaspora Mapping: 8 juta diaspora Indonesia di 50+ negara, 500rb profesional maritime/biotech.
- Engagement Model: MERLIN Diaspora Network (MDN), chapter per country, digital platform.
- Diaspora Contribution: Investment (Rp50 triliun target), expertise (mentor, advisor), market access (B2B network), advocacy.
- Programs: Diaspora investment fund, diaspora talent program, diaspora market access, diaspora homecoming.
- Incentive: Tax holiday for diaspora investment, dual citizenship consideration, property right, recognition program.
- Success Metrics: 100.000 diaspora member, Rp50 triliun diaspora investment, 1000 diaspora talent placement.

Owner: Diaspora Director MERLIN

PGA-79: Geopolitical Risk Radar

Tujuan: Radar risiko geopolitik MERLIN.

Elemen Kunci:

- Risk Monitoring: 30+ geopolitical risk - US-China trade war, South China Sea, Taiwan strait, Middle East, Russia-Ukraine, climate treaty, ocean governance.
- Risk Scenarios: 5 skenario per risk with probability, impact, mitigation.
- Early Warning: Subscription to geopolitical intelligence (Eurasia Group, Stratfor), expert network, dashboard.
- Impact Assessment: Supply chain, market access, investment, regulatory, currency, security.
- Response Playbook: Per risk scenario, response option, decision framework, escalation.
- Review: Monthly risk radar, quarterly deep dive, annual scenario planning.

Owner: Chief Risk + Diplomatic Officer

PGA-80: Trade Treaty Compliance Atlas

Tujuan: Atlas kepatuhan perjanjian dagang MERLIN.

Elemen Kunci:

- Trade Treaties: WTO, ASEAN FTA, RCEP, Indonesia-EU CEPA (negotiation), Indonesia-Japan IJEPA, Indonesia-Korea IKCEPA, Indonesia-Australia IA-CEPA, GSP.
- Compliance per Treaty: Rules of origin, tariff schedule, SPS measure, TBT, customs procedure, IP, dispute.
- Utilization Strategy: Maximize FTA preference, tariff engineering, supply chain optimization.

- Compliance Team: 10 trade compliance specialist, customs broker partnership, legal counsel.
- Documentation: Certificate of origin, FTA form, customs declaration, supporting document.
- Audit: Annual trade compliance audit, customs audit response, FTA utilization review.

Owner: Trade Compliance Director MERLIN

◆ Akhir Domain 10 — 8 Dokumen PGA ◆

DOMAIN 11 | SOVEREIGNTY ECONOMICS & ISLAMIC FINANCE

Ekonomi Kedaulatan & Keuangan Syariah MERLIN

Range: PGA-81 s/d PGA-88 | Total: 8 Dokumen

PGA-81: Syariah Compliance Framework MERLIN

Tujuan: *Kerangka kepatuhan syariah MERLIN.*

Elemen Kunci:

- Syariah Principles: Halal (pure), Toyibban (good), no riba (interest), no gharar (uncertainty), no maysir (gambling), no haram activity.
- Syariah Board: Dewan Pengawas Syariah (DPS) 7 ulama + economist, fatwa per product/transaction.
- Halal Certification: MUI Halal for all food/cosmetic products, BPJPH registration, JAKIM (Malaysia), HCA (global).
- Syariah Finance: Sukuk, Islamic bank (BSI, Muamalat), Islamic insurance (takaful), Islamic investment (Reksa Dana Syariah).
- Syariah Audit: Annual syariah audit, compliance report, fatwa tracker, remediation.
- Syariah Training: Annual training for all staff, halal awareness, syariah finance literacy.

Owner: *Dewan Pengawas Syariah MERLIN*

PGA-82: Dewan Pengawas Syariah Charter

Tujuan: *Piagam Dewan Pengawas Syariah MERLIN.*

Elemen Kunci:

- Mandate: Ensure all MERLIN activity is syariah-compliant, issue fatwa, audit compliance.
- Composition: 7 members - 4 ulama (maritime, economy, finance, food), 3 economist (Islamic finance, halal industry, ESG).
- Appointment: By Dewan Pembina, term 3 years renewable, independence guaranteed.
- Meeting: Monthly meeting, emergency fatwa 24 hours, annual syariah review.
- Fatwa Process: Question → Research → Deliberation → Fatwa → Documentation → Communication.
- Authority: Veto on non-syariah activity, mandatory syariah audit, public report.

Owner: *Ketua Dewan Pengawas Syariah*

PGA-83: Zakat & Infaq Management Policy

Tujuan: *Kebijakan manajemen zakat dan infaq MERLIN.*

Elemen Kunci:

- Zakat Sources: Zakat mal (2,5% asset), zakat fitrah (Ramadan), infaq (voluntary), sedekah (voluntary).
- Zakat Collection: BAZNAS partnership, LAZ MERLIN (own amil), digital collection (e-wallet, bank transfer).
- Zakat Distribution: 8 asnaf (fakir, miskin, amil, mualaf, riqab, gharimin, fisabilillah, Ibnu Sabil). Priority to member petani.
- Programs: Beasiswa anak petani, modal usaha petani, healthcare petani, emergency relief, dakwah maritime.
- Target: Zakat collection Rp1 triliun/tahun by 2030, distribution 95% within 12 months.
- Reporting: Annual zakat report, audit BAZNAS, transparency dashboard, donor receipt.

Owner: LAZ MERLIN (Lembaga Amil Zakat)

PGA-84: Wakaf Produktif Endowment Model

Tujuan: Model wakaf produktif MERLIN sebagai endowment abadi.

Elemen Kunci:

- Wakaf Concept: Nazhir (MERLIN), maukuf 'alaih (member petani), mauquf (asset). Perpetual endowment.
- Wakaf Asset: Tanah budidaya (10.000 ha), bangunan EBIP (40 unit), saham (10% MERLIN), cash endowment (Rp10 triliun).
- Productive Model: Asset dioperasikan → hasil untuk program (member welfare, riset, dakwah).
- Wakaf Programs: Wakaf bibit (perpetual supply), wakaf scholarship (1000 anak petani/tahun), wakaf healthcare (klinik gratis), wakaf riset (5% revenue).
- Nazhir Governance: BWI registration, annual report, audit, transparency.
- Target: Wakaf Rp50 triliun by 2030, 100.000 wakif, income Rp5 triliun/tahun untuk program.

Owner: Nazhir Wakaf MERLIN

PGA-85: Sukuk Issuance Framework

Tujuan: Kerangka penerbitan sukuk MERLIN.

Elemen Kunci:

- Sukuk Concept: Islamic bond berbasis aset, no riba, sesuai syariah. Alternative to conventional bond.
- Sukuk Structure: Ijarah (lease), Mudharabah (profit sharing), Musyarakah (partnership), Wakalah (agency).
- Issuance Plan: Sukuk MERLIN 2027 Rp50 triliun (Series A), 2028 Rp100 triliun (Series B), 2030 Rp200 triliun (Series C).
- Underlying Asset: EBIP (40 unit), Blue Tourism (40 unit), pengolahan rumput laut.
- Investor: Domestic (BSI, Muamalat, pension syariah), International (Islamic bank Gulf, SE Asia).
- Listing: Bursa Efek Indonesia (IDX), future Luxembourg Stock Exchange for global sukuk.

- Rating: AAA target (syariah rating), ESG rating, sovereign guarantee consideration.

Owner: CFO + Syariah Board MERLIN

PGA-86: Seed & Food Sovereignty Declaration

Tujuan: Deklarasi kedaulatan benih dan pangan MERLIN.

Elemen Kunci:

- Seed Sovereignty: Bibit rumput laut unggul dikembangkan, dimiliki, dan didistribusikan oleh Indonesia. No foreign monopoly.
- Bibit MERLIN: 10 varietas unggul (4 Kappaphycus, 3 Eucheuma, 3 Gracilaria) - tahan penyakit, cepat tumbuh, kualitas tinggi.
- Bibit R&D: 5% revenue untuk R&D bibit, 100 peneliti, kemitraan dengan 20 universitas, 1000+ aksi genetik.
- Bibit Distribution: MERLIN Seed Center per kabupaten (350 unit), free bibit for member, quality guaranteed.
- Food Sovereignty: Rumput laut sebagai sumber pangan alternatif (protein, vitamin, mineral), fortifikasi pangan.
- Food Products: 50+ food products (noodle, snack, beverage, supplement), fortifikasi tepung, roti, mi.

Owner: Chief Seed & Food Sovereignty Officer

PGA-87: Energy Sovereignty Blueprint

Tujuan: Cetak biru kedaulatan energi MERLIN.

Elemen Kunci:

- Bioethanol E30: 1 juta kiloliter/tahun dari rumput laut, substitusi premium, mengurangi impor minyak.
- Biogas: 1 MW per EBIP $\times 40 = 40$ MW, untuk operasional EBIP, surplus ke grid.
- Biochar: 100.000 ton/tahun, untuk pupuk + carbon sequestration.
- Green Hydrogen: Pilot 2028, scale 2030 10.000 ton/tahun, for export to Japan/Korea.
- Renewable Mix: EBIP 100% renewable (biogas + solar + green hydrogen), surplus to grid.
- Energy Independence: MERLIN self-sufficient energy, surplus for community, reduce fossil import.

Owner: Chief Energy Officer MERLIN

PGA-88: Agrarian Reform & Land Justice

Tujuan: Reforma agraria dan keadilan lahan untuk member MERLIN.

Elemen Kunci:

- Land Rights: Petani mendapat hak kelola lahan laut (HGU komunal, HKM, HPA). Tidak ada tengkulak yang menguasai.

- Agrarian Reform: 1 juta ha laut dialihkan ke koperasi member, dari korporasi/tengkulak.
- Land Registration: Sertifikat tanah laut per kelompok tani, digitized, transparan.
- Land Conflict Resolution: Mediasi, pengadilan agraria, restorative justice.
- Land Restoration: Laut terdegradasi direstorasi (coral reef, mangrove, seagrass).
- Land Justice: Petani adat, perempuan, muda mendapat prioritas akses lahan. No discrimination.

Owner: Chief Agrarian Officer MERLIN

◆ Akhir Domain 11 — 8 Dokumen PGA ◆

DOMAIN 12 | SOCIAL JUSTICE, EQUITY & HUMAN RIGHTS

Keadilan Sosial, Kesetaraan & Hak Asasi MERLIN

Range: PGA-89 s/d PGA-96 | Total: 8 Dokumen

PGA-89: Women's Economic Justice Charter

Tujuan: *Piagam keadilan ekonomi perempuan MERLIN.*

Elemen Kunci:

- Prinsip: 50% perempuan di seluruh level, equal pay for equal work, leadership pipeline perempuan.
- Programs: Women Entrepreneur MERLIN (100.000 UMKM perempuan), Women Leadership Academy, childcare center di EBIP, maternal protection.
- Perempuan Petani: 60% petani rumput laut adalah perempuan. MERLIN pastikan harga jaminan setara, akses pembayaran langsung ke perempuan.
- Perempuan Leader: 50% manager, 40% director, 30% board perempuan by 2030.
- Protection: Anti-harassment, anti-discrimination, maternity leave 6 bulan, menstrual leave 2 hari/bulan.
- Success Metrics: 50% women workforce, 30% women board, 100.000 women entrepreneur, 0 harassment case.

Owner: *Chief Diversity & Inclusion Officer*

PGA-90: Youth Parliament Constitution

Tujuan: *Konstitusi Parlemen Pemuda MERLIN.*

Elemen Kunci:

- Concept: Parlemen pemuda sebagai suara 1 juta pemuda di ekosistem MERLIN.
- Composition: 100 anggota (10 per regional), usia 18-35, dipilih 2 tahun.
- Mandate: Advokasi kepentingan pemuda, program pemuda, pengawasan implementasi.
- Programs: Youth scholarship, Youth entrepreneurship, Youth leadership academy, Youth festival.
- Budget: 5% MERLIN revenue untuk program pemuda, dikelola parlemen pemuda.
- Success Metrics: 1 juta pemuda engaged, 100.000 entrepreneur pemuda, 1000 leader pemuda.

Owner: *Ketua Parlemen Pemuda MERLIN*

PGA-91: Disability & Universal Access Policy

Tujuan: *Kebijakan disability dan akses universal MERLIN.*

Elemen Kunci:

- Prinsip: 5% tenaga kerja adalah disability, universal access di seluruh fasilitas, equal opportunity.

- Universal Design: EBIP, Blue Tourism, kantor, transportasi, digital platform - accessible untuk disability.
- Programs: Disability skill training, workplace accommodation, assistive technology, disability entrepreneur.
- Accessibility: Ramp, elevator, braille signage, sign language interpreter, accessible website (WCAG 2.1 AA).
- Protection: Anti-discrimination, reasonable accommodation, grievance mechanism.
- Success Metrics: 5% disability workforce, 100% accessible facility, 1000 disability entrepreneur.

Owner: Chief Accessibility Officer MERLIN

PGA-92: Indigenous Knowledge Protection

Tujuan: *Perlindungan pengetahuan adat MERLIN.*

Elemen Kunci:

- Indigenous Knowledge: Praktik budidaya tradisional, pengetahuan laut lokal, ritual, obat-obatan tradisional.
- Protection: Documentation, prior informed consent, benefit sharing, IP protection (collective trademark, geographical indication).
- Programs: Indigenous Knowledge Database, elder interview, community documentation, cultural center.
- Benefit Sharing: Royalty dari komersialisasi, community development fund, scholarship for indigenous youth.
- Indigenous Member: Prioritas penerimaan member adat, hak kelola wilayah adat, consultation protocol.
- Success Metrics: 1000 indigenous knowledge documented, 50 geographical indication, 100 community fund.

Owner: Chief Indigenous Affairs Officer MERLIN

PGA-93: Child Protection & Safeguarding

Tujuan: *Perlindungan anak dan safeguarding MERLIN.*

Elemen Kunci:

- Child Protection Policy: Zero tolerance for child labor, child abuse, child exploitation. CRC compliance.
- Child Labor: No child <15 in any MERLIN activity. Children 15-17 only light work with school priority.
- Education: 100% anak member sekolah, scholarship, free school supplies, after-school program.
- Healthcare: Free healthcare for children member, immunization, nutrition, mental health.
- Child Safeguarding: Background check all staff, child-safe code of conduct, reporting mechanism.
- Programs: Anak Bahari scholarship, Summer camp, Marine education, Talent development.

Owner: Chief Child Protection Officer MERLIN

PGA-94: Climate Justice & Adaptation

Tujuan: Keadilan iklim dan adaptasi MERLIN.

Elemen Kunci:

- Climate Justice: Petani pesisir adalah korban iklim, berhak atas adaptation fund, loss & damage compensation.
- Adaptation Program: Bibit tahan stress (heat, salinity, acid), diversifikasi lokasi, early warning system, insurance.
- Mitigation: 12 juta ton CO2/tahun terserap, bioethanol E30 substitusi fosil, bioplastic substitusi plastic.
- Climate Finance: Adaptation fund USD 500 juta/tahun, Green Climate Fund, loss & damage fund.
- Just Transition: Petani yang terdampak iklim direlokasi, dilatih ulang, diberi modal.
- Success Metrics: 100% member climate-resilient, 12 juta ton CO2, USD 500 juta climate finance, 0 climate displacement.

Owner: Chief Climate Justice Officer MERLIN

PGA-95: Migrant Worker Integration

Tujuan: Integrasi pekerja migran MERLIN.

Elemen Kunci:

- Migrant Worker: 5 juta TKI di luar negeri, banyak dari pesisir. MERLIN facilitate return & reintegration.
- Programs: Pre-departure training (seaweed skill), remittance channeling to MERLIN investment, returnee reintegration.
- Returnee: Skill certification, entrepreneurship, job placement in MERLIN EBIP.
- Family Support: Family left behind dapat member benefits, scholarship for children, healthcare.
- Remittance: MERLIN Remittance service, lower fee, channeling to productive investment.
- Success Metrics: 100.000 returnee integrated, Rp10 triliun remittance channeled, 500.000 family supported.

Owner: Migrant Worker Director MERLIN

PGA-96: Food Justice Manifesto

Tujuan: Manifesto keadilan pangan MERLIN.

Elemen Kunci:

- Food as Human Right: Setiap orang berhak atas pangan cukup, bergizi, aman, halal. MERLIN pastikan.
- Food Security: Rumput laut sebagai sumber pangan berkelanjutan, fortifikasi pangan lokal, distribusi merata.

- Food Access: Harga terjangkau untuk member, subsidi untuk miskin, food bank untuk darurat.
- Food Quality: Halal, organic, bebas kontaminan, gizi seimbang, food safety standard.
- Food Sovereignty: Indonesia mandiri pangan, tidak impor, tidak ketergantungan asing.
- Programs: Rumput Laut for School Feeding, fortifikasi tepung, subsidized food for poor, food bank.

Owner: Chief Food Justice Officer MERLIN

◆ Akhir Domain 12 — 8 Dokumen PGA ◆

DOMAIN 13 | FUTURE CIVILIZATION & FRONTIER SCIENCE

Peradaban Masa Depan & Ilmu Perintis MERLIN

Range: PGA-97 s/d PGA-108 (12 dokumen) | Total: 12 Dokumen

PGA-97: AI Ethics & Algorithmic Sovereignty MERLIN

Tujuan: *Etika AI dan kedaulatan algoritmik MERLIN.*

Elemen Kunci:

- AI Principles: Fairness, transparency, accountability, privacy, safety, human-in-the-loop, sovereignty.
- AI Applications: Aquaculture monitoring, quality inspection, supply chain optimization, carbon MRV, member service.
- Algorithmic Sovereignty: AI model dikembangkan, dimiliki, dioperasikan oleh MERLIN. No foreign dependency for critical AI.
- MERLIN AI Lab: 100 AI engineer, partnership with 10 university, GPU cluster, MLOps platform.
- AI Governance: AI Ethics Board, algorithm audit, bias detection, explainability, recourse mechanism.
- Open Source: Non-critical AI model open-sourced for community benefit, critical model proprietary.

Owner: *Chief AI Officer MERLIN*

PGA-98: Quantum Computing Transition MERLIN

Tujuan: *Transisi komputasi kuantum MERLIN.*

Elemen Kunci:

- Quantum Use Cases: Material design (seaweed polymer), optimization (logistics), drug discovery (bioactive compound), cryptography.
- Quantum Readiness: Partnership with IBM Quantum, Google Quantum, startup (Xanadu, IonQ), 5-year roadmap.
- Quantum Lab: 1 quantum researcher, 1 quantum engineer, access to quantum cloud (IBM, AWS Braket).
- Quantum Cryptography: Post-quantum cryptography migration, QKD pilot for critical communication.
- Quantum Talent: 10 PhD scholarship in quantum, 100 staff trained quantum literacy, university partnership.
- Timeline: 2026-2028 quantum literacy, 2028-2030 quantum pilot, 2030-2035 quantum production.

Owner: *Chief Quantum Officer MERLIN*

PGA-99: Space & Satellite Economy Strategy

Tujuan: *Strategi ekonomi luar angkasa dan satelit MERLIN.*

Elemen Kunci:

- Satellite Application: Earth observation (ocean monitoring, illegal fishing, climate), communication (maritime broadband), navigation (precision aquaculture).
- Partnership: BRIN LAPAN, SpaceX Starlink, OneWeb, regional satellite (Garuda).
- MERLIN Satellite: Future own satellite (MERLIN-Sat) for exclusive monitoring, 2032 launch.
- Data Analytics: Satellite imagery for site selection, growth monitoring, harvest prediction, climate modeling.
- Space Economy: Microgravity biotech (future), space-based solar (future), lunar mining (far future).
- Budget: Rp500 miliar 2026-2030 for satellite data and partnership, Rp2 triliun for MERLIN-Sat 2032.

Owner: *Chief Space Officer MERLIN*

PGA-100: Bioeconomy & Synthetic Biology Policy

Tujuan: *Kebijakan bioekonomi dan biologi sintetik MERLIN.*

Elemen Kunci:

- Bioeconomy: Rumput laut sebagai basis bioekonomi - biomaterial, bioenergy, biochemical, biopharma.
- Synthetic Biology: Engineered microbe for efficient extraction, novel compound synthesis, custom enzyme.
- Biorefinery: Multi-product dari rumput laut (food, fuel, material, chemical), zero waste biorefinery.
- Biosafety: Containment, biosecurity, ethical use, Cartagena Protocol compliance.
- IP & Benefit Sharing: Paten for novel compound, Nagoya Protocol for genetic resource, community benefit.
- Lab Infrastructure: BSL-2 lab per EBIP, BSL-3 lab di research center, biofoundry for synbio.

Owner: *Chief Biotech Officer MERLIN*

PGA-101: Open Source & Digital Commons

Tujuan: *Open source dan digital commons MERLIN.*

Elemen Kunci:

- Open Source: Non-critical software open-sourced (MERLIN OS, MERLIN Aquaculture AI, MERLIN Carbon MRV).
- Digital Commons: Public dataset (ocean data, seaweed genome, market data), public good.
- Open Hardware: Low-cost aquaculture sensor design open, 3D-printable, community-built.

- Open Knowledge: Research publication open access, training material open, best practice shared.
- Community: Developer community, contributor program, hackathon, bounty program.
- Impact: 100 open source project, 1000 contributor, 10 juta user, multiplier effect.

Owner: Chief Open Innovation Officer MERLIN

PGA-102: Health Data & Genomics Sovereignty

Tujuan: Kedaulatan data kesehatan dan genomik MERLIN.

Elemen Kunci:

- Health Data: Member health record, telemedicine, wearable data, environmental exposure.
- Genomics: Seaweed genome (1000+ accessions), human genome (member consent), microbiome.
- Sovereignty: Data stored in Indonesia, no foreign transfer without consent, sovereign cloud.
- MERLIN Genome Lab: Whole genome sequencing, bioinformatics, gene editing (CRISPR for seaweed).
- Personalized Health: Nutrition genomics, precision medicine, biomarker for member health.
- Ethics: Informed consent, data minimization, anonymization, benefit sharing, anti-discrimination.

Owner: Chief Genomics Officer MERLIN

PGA-103: Ocean & Blue Economy Sovereignty

Tujuan: Kedaulatan laut dan ekonomi biru MERLIN.

Elemen Kunci:

- Ocean Sovereignty: Wawasan nusantara, EEZ 6,4 juta km², territorial sea 3,1 juta km², coastline 108.000 km.
- Blue Economy: Sustainable ocean economic - fishery, aquaculture, energy, tourism, biotech, shipping, carbon.
- MERLIN Role: Seaweed as flagship blue economy, integrated model, sovereign control.
- Ocean Governance: UNCLOS, regional cooperation (ASEAN, APEC, PIF), IUU fishing combat.
- Marine Spatial Planning: Zoning for aquaculture, fishing, tourism, conservation, energy.
- Marine Protection: 30% marine protected area by 2030, restoration, blue carbon ecosystem.

Owner: Chief Ocean Sovereignty Officer MERLIN

PGA-104: Circular Economy Certification

Tujuan: Sertifikasi ekonomi sirkular MERLIN.

Elemen Kunci:

- Circular Principles: Reduce, reuse, recycle, refuse, rot. Zero waste, closed-loop, regenerative.
- MERLIN Circular Model: Rumput laut 100% utilized - main product (60%), byproduct (30%), waste-to-energy (8%), compost (2%).

- Byproduct Utilization: Sludge → fertilizer, wash water → biogas, residue → biochar, packaging → bioplastic.
- Certification: Cradle to Cradle, EU Ecolabel, ISO 14021 (recycled content), MERLIN Zero Waste Certified.
- Standard Setting: MERLIN buat standar nasional Zero Waste Seaweed Industry, kriterianya meliputi: 0% landfill, 95% resource utilization, 100% renewable energy.
- Audit: Annual circular audit, third-party verification, public disclosure.

Owner: Chief Circular Economy Officer MERLIN

PGA-105: Carbon Market & Blue Carbon Strategy

Tujuan: Strategi pasar karbon dan blue carbon MERLIN.

Elemen Kunci:

- Blue Carbon Source: Seaweed (12 juta ton CO₂/tahun), mangrove (restorasi 100rb ha), seagrass (restorasi 50rb ha).
- Carbon Credit: Verra VCS (VM0033 seaweed), Gold Standard, Plan Vivo. 12 juta credit/tahun.
- Carbon Market: Compliance market (EU ETS, Indonesia ICCP), voluntary market (corporate offset).
- Pricing: USD 30-80/ton CO₂, premium for blue carbon, projected USD 50-150 by 2035.
- Revenue: 12 juta × USD 50 = USD 600 juta/year (Rp9 triliun), 10% to member, 90% to ecosystem.
- MRV: Satellite + IoT + manual measurement, blockchain registry, third-party verification.
- Beyond Carbon: Biodiversity credit, plastic credit, water credit - future revenue stream.

Owner: Chief Carbon Officer MERLIN

PGA-106: Crisis Simulation War Gaming Protocol

Tujuan: Protokol simulasi krisis dan war gaming MERLIN.

Elemen Kunci:

- War Gaming: Simulation of crisis scenario - climate disaster, market crash, supply chain disruption, cyber attack, pandemic.
- Scenario Library: 20+ scenario with script, role, injection, decision point, debrief.
- Frequency: Annual full simulation (2 days), quarterly tabletop (4 hours), monthly component.
- Participants: Executive, manager, operator, partner, regulator, community.
- Outcome: Decision-making capability, system resilience, gap identification, improvement plan.
- Documentation: After-action report, lesson learned, action item, follow-up tracking.

Owner: Chief Resilience Officer MERLIN

PGA-107: Interoperability Standards Atlas

Tujuan: Atlas standar interoperabilitas MERLIN.

Elemen Kunci:

- Interoperability Layers: Technical (API, protocol), Semantic (data model, ontology), Organizational (governance, process), Legal (agreement, liability).
- Standards: ISO 19444 (interoperability), GS1 (traceability), UN/CEFACT (trade), OGC (geospatial), HL7 (health).
- MERLIN Standards: Seaweed data model, traceability schema, carbon MRV protocol, member ID system.
- Open API: 100+ REST/GraphQL API, sandbox, documentation, developer portal.
- Data Exchange: EDI, API, file-based, real-time streaming. With partner, regulator, customer.
- Certification: Interoperability certification per system, annual recertification, compliance audit.

Owner: Chief Interoperability Officer MERLIN

PGA-108: Perisai Konstitusional 100 Tahun MERLIN

Tujuan: Perisai konstitusional 100 tahun MERLIN.

Elemen Kunci:

- Constitutional Shield: MERLIN dilindungi konstitusi (UU Rumput Laut), tidak mudah dibubarkan/diubah pemerintahan.
- 100-Year Framework: Berdiri 2026, 100 tahun 2126. Tidak boleh dibubarkan kecuali musyawarah nasional 2/3 member.
- Endowment: Rp50 triliun endowment fund, income abadi untuk operasional & program.
- Asset Lock: Aset MERLIN tidak boleh dijual/dibagi kecuali untuk ekosistem rumput laut Indonesia.
- Constitutional Amendment: Hanya melalui Musda 2/3 member + approval Dewan Pembina + Mahkamah Konstitusi.
- International Treaty: MERLIN registered dengan UN, diakui sebagai public good organization.

Owner: Dewan Pembina + Legal MERLIN

◆ Akhir Domain 13 — 12 Dokumen PGA ◆

DOMAIN 14 | DEFENSE, SECURITY & RESILIENCE

Pertahanan, Keamanan & Ketahanan MERLIN

Range: PGA-109 s/d PGA-116 | Total: 8 Dokumen

PGA-109: National Security Doctrine MERLIN

Tujuan: *Doktrin keamanan nasional MERLIN.*

Elemen Kunci:

- Security Philosophy: Economic security is national security. MERLIN as economic defense line.
- Threat Landscape: Economic (competitor, market manipulation), Environmental (climate, pollution), Cyber (hack, data breach), Physical (illegal fishing, sabotage), Social (disinformation, conflict).
- Defense in Depth: Prevention, detection, response, recovery, learning. Multi-layer defense.
- Partnership: TNI AL, Polairud, Bakamla, KKP, Intelijen. Civil-military cooperation.
- Self-Defense Capability: MERLIN Security Force (10.000 personel), patrol boat, drone, CCTV, AI surveillance.
- Resilience: Redundancy, diversification, stockpiling, mutual aid, business continuity.

Owner: *Chief Security Officer MERLIN*

PGA-110: Cybersecurity Master Plan

Tujuan: *Master plan keamanan siber MERLIN.*

Elemen Kunci:

- Cyber Threats: Ransomware, phishing, supply chain attack, insider, APT, DDoS, IoT attack.
- Defense Framework: NIST CSF - Identify, Protect, Detect, Respond, Recover.
- Security Stack: Firewall, IDS/IPS, SIEM, EDR, DLP, IAM, MFA, encryption, backup.
- Security Operations: SOC 24/7, threat hunting, incident response, forensics, threat intel.
- Compliance: ISO 27001, SOC 2, PCI DSS, UU PDP, GDPR. Annual certification.
- Training: Annual security awareness, phishing simulation, role-specific training, secure development.
- Investment: Rp500 miliar 2026-2030, 2% IT budget, security-by-design.

Owner: *CISO MERLIN*

PGA-111: Counter-Intelligence Protocol

Tujuan: *Protokol kontra-intelijen MERLIN.*

Elemen Kunci:

- Threat: Industrial espionage, foreign intelligence, competitor theft, insider threat.

- Counter-Intel: Background check, security clearance, need-to-know, travel protocol, foreign engagement protocol.
- Insider Threat: Behavior analytics, whistleblower, periodic reinvestigation, exit interview.
- Foreign Engagement: Disclosure protocol, partnership due diligence, technology transfer control.
- Information Security: Classification, marking, handling, disposal, audit.
- Coordination: BIN, BAIS TNI, intelijen KKP. Information sharing on threat.

Owner: Chief Counter-Intel Officer MERLIN

PGA-112: Emergency Response Framework

Tujuan: Kerangka respons darurat MERLIN.

Elemen Kunci:

- Emergency Types: Natural (tsunami, earthquake, storm), Industrial (fire, chemical spill, explosion), Health (pandemic, food poisoning), Security (attack, hijack, kidnap), Cyber (breach, ransomware).
- Response Framework: Command (EOC), Operations (responder), Planning (intel), Logistics (supply), Finance (cost).
- Emergency Operations Center (EOC): 1 pusat Jakarta, 4 regional, 40 EBIP. 24/7 activation.
- Response Protocol: Alert → Assess → Activate → Respond → Recover → Review. SLA per severity.
- Mutual Aid: Agreement with TNI, PMI, BASARNAS, local government. Resource sharing.
- Communication: Mass notification, public information, media, family liaison.

Owner: Chief Emergency Officer MERLIN

PGA-113: Disaster Management Protocol

Tujuan: Protokol manajemen bencana MERLIN.

Elemen Kunci:

- Disaster Cycle: Mitigation → Preparedness → Response → Recovery → Reconstruction.
- Disaster Types: Tsunami, earthquake, volcanic, flood, drought, El Nino, La Nina, oil spill, harmful algal bloom.
- Early Warning: BMMD, BMKG integration, IoT sensor network, satellite, community warning.
- Preparedness: Evacuation plan, shelter, stockpile (food, water, medical), drill annual.
- Response: Search & rescue, medical, shelter, food, water, communication, security.
- Recovery: Damage assessment, reconstruction (build back better), livelihood recovery, mental health.
- Insurance: Catastrophe insurance, micro-insurance for member, reinsurance pool.

Owner: Chief Disaster Officer MERLIN

PGA-114: Business Continuity Plan

Tujuan: *Rencana kontinuitas bisnis MERLIN.*

Elemen Kunci:

- BCP Framework: ISO 22301. Business Impact Analysis, Risk Assessment, Strategy, Plan, Test, Maintain.
- Critical Functions: Production, supply chain, payment, communication, member service, healthcare.
- RTO/RPO: Production 7 hari/1 hari, Payment 1 hari/4 jam, Communication 4 jam/1 jam, Healthcare 1 jam/real-time.
- Continuity Strategies: Redundancy (backup site, equipment), Diversification (supplier, location), Insurance, Mutual aid.
- Crisis Communication: Internal (cascade in 1 hour), External (media in 4 hours), Member (SMS in 2 hours).
- Testing: Annual full BCP test, semi-annual tabletop, quarterly component. Update after test.

Owner: *Chief Business Continuity Officer MERLIN*

PGA-115: Crisis Communication Protocol

Tujuan: *Protokol komunikasi krisis MERLIN.*

Elemen Kunci:

- Crisis Team: CEO, COO, CCO (Chief Comms), CLO (Legal), CHRO, CISO. War room activation.
- Response Timeline: Detection → Verification (1 hour) → Holding statement (2 hours) → Full response (8 hours) → Update (daily).
- Communication Channels: Press release, press conference, social media, member SMS, intranet, direct email.
- Stakeholder Priority: Victim & family > Member > Employee > Regulator > Customer > Media > Public.
- Message Principles: Truthful, timely, empathetic, action-oriented, transparent.
- Recovery: Post-crisis review, reputation repair, lesson learned, prevention action.

Owner: *Chief Communications Officer MERLIN*

PGA-116: Resilience & Adaptation Framework

Tujuan: *Kerangka ketahanan dan adaptasi MERLIN.*

Elemen Kunci:

- Resilience Dimensions: Economic, Operational, Social, Environmental, Institutional, Technological.
- Adaptation Pathway: Monitor → Anticipate → Adapt → Transform. Continuous evolution.
- Stress Test: Annual stress test - financial, operational, climate, cyber, social.
- Adaptive Capacity: Diversity, redundancy, modularity, flexibility, learning.
- Transformation: 5-year strategic review, 10-year reinvention, 25-year transformation.

- Indicators: Resilience index (composite), recovery time, adaptation cost, transformation success.

Owner: Chief Resilience Officer MERLIN

◆ Akhir Domain 14 — 8 Dokumen PGA ◆

DOMAIN 15 | TECHNOLOGY, INNOVATION & DIGITAL SOVEREIGNTY

Teknologi, Inovasi & Kedaulatan Digital MERLIN

Range: PGA-117 s/d PGA-124 | Total: 8 Dokumen

PGA-117: Digital Transformation Roadmap

Tujuan: *Peta jalan transformasi digital MERLIN.*

Elemen Kunci:

- Digital Vision: MERLIN sebagai most digitally-advanced seaweed ecosystem. Paperless, contactless, AI-first.
- Pillar 1 - Core Digital: ERP (SAP), CRM (Salesforce), SCM, HRIS, Finance - integrated platform.
- Pillar 2 - Industry 4.0: IoT aquaculture, smart factory, robotics, digital twin per EBIP.
- Pillar 3 - Member Digital: Mobile app (member portal, marketplace, telemedicine, e-learning, fintech).
- Pillar 4 - Data & AI: Data lake, ML platform, AI applications, analytics, dashboard.
- Pillar 5 - Digital Sovereign: MERLIN Cloud, MERLIN OS, open source, data sovereignty.
- Investment: Rp5 triliun 2026-2030, 500 IT staff, partnership with hyperscaler (AWS, Azure, GCP).

Owner: *CDO + CTO MERLIN*

PGA-118: AI & Automation Strategy

Tujuan: *Strategi AI dan otomasi MERLIN.*

Elemen Kunci:

- AI Applications: Aquaculture monitoring (image recognition, growth prediction), Quality inspection (computer vision), Supply chain optimization (forecasting), Member service (chatbot, personalization), Carbon MRV (satellite AI).
- Automation: Smart factory (robotic processing), Warehouse (AGV), Logistics (route optimization), Admin (RPA), Member service (AI agent).
- MERLIN AI Lab: 100 AI engineer, 10 PhD, partnership with 10 university, GPU cluster, MLOps.
- AI Ethics: Fairness, transparency, accountability, no bias, human-in-the-loop, recourse.
- AI Governance: AI Board, algorithm audit, model registry, version control, monitoring.
- Impact: 30% productivity gain, 20% cost reduction, 50% faster decision, 100 new AI products.

Owner: *Chief AI Officer MERLIN*

PGA-119: Blockchain & Decentralized Systems

Tujuan: *Blockchain dan sistem terdesentralisasi MERLIN.*

Elemen Kunci:

- Blockchain Use Cases: Traceability (farm to shelf), Carbon credit registry, Member ID & credential, Smart contract (offtake, payment), Token (loyalty, reward).
- Platform: Hyperledger Fabric (permissioned for enterprise), Polygon (public for carbon), IPFS (storage).
- MERLIN Chain: Future own blockchain for ecosystem, EVM-compatible, low fee, scalable.
- Traceability: QR code per batch, blockchain record of every step (seed, harvest, process, ship), consumer scan.
- Carbon Registry: Each ton carbon = 1 NFT, transparent registry, retired tracking, double-counting prevention.
- Smart Contract: Offtake agreement auto-execute, payment on delivery, penalty on default.
- DAO: Future MERLIN DAO for member governance, voting on key issues, treasury management.

Owner: Chief Blockchain Officer MERLIN

PGA-120: Cloud Infrastructure Policy

Tujuan: Kebijakan infrastruktur cloud MERLIN.

Elemen Kunci:

- Cloud Strategy: Hybrid cloud - public cloud for non-critical, private cloud for critical, edge for IoT.
- Public Cloud: AWS, Azure, GCP - multi-cloud for vendor diversification, residency in Indonesia region.
- Private Cloud: MERLIN Cloud - on-prem for critical data (member data, IP, financial), OpenStack, sovereign.
- Edge Computing: Edge server per EBIP for real-time IoT processing, low latency, offline capability.
- Data Residency: All member data in Indonesia, no cross-border without consent, encryption everywhere.
- Cloud Governance: Cloud Center of Excellence, FinOps (cost optimization), Security (CSPM, CWPP), Architecture review.
- Investment: Rp1 triliun/year cloud spend, 50 cloud engineer, multi-year commitment discount.

Owner: Chief Cloud Officer MERLIN

PGA-121: Software Development Standards

Tujuan: Standar pengembangan perangkat lunak MERLIN.

Elemen Kunci:

- SDLC: Agile (Scrum/Kanban), 2-week sprint, CI/CD, DevOps, DevSecOps.
- Tech Stack: Backend (Java Spring, Go, Python FastAPI), Frontend (React, Next.js, React Native), Data (PostgreSQL, MongoDB, Snowflake), AI (TensorFlow, PyTorch).

- Code Quality: Code review mandatory, static analysis (SonarQube), unit test 80% coverage, integration test.
- Security: SAST, DAST, SCA, IaC scanning, secret management, secure coding training.
- Documentation: README, API doc (OpenAPI), architecture doc, runbook.
- Release Management: Trunk-based development, feature flag, blue-green deploy, canary release.
- Observability: Logging (ELK), Monitoring (Prometheus, Grafana), Tracing (Jaeger), APM (Datadog).

Owner: CTO + VP Engineering MERLIN

PGA-122: Data Privacy & Protection

Tujuan: Privasi dan perlindungan data MERLIN.

Elemen Kunci:

- Compliance: UU PDP Indonesia (effective 2024), GDPR (for EU customer), CCPA (for US customer), ISO 27701.
- Privacy Principles: Lawfulness, fairness, transparency, purpose limitation, data minimization, accuracy, storage limitation, integrity, accountability.
- Privacy by Design: Default privacy, DPIA (Data Protection Impact Assessment), privacy engineering.
- Member Rights: Access, rectification, erasure, portability, objection, automated decision.
- Data Breach: Notification within 3x24 jam to member & regulator, breach response plan, forensics.
- DPO: Data Protection Officer, independent, direct report to board, certification (CIPP, CIPM).

Owner: DPO MERLIN

PGA-123: Technology Acquisition Framework

Tujuan: Kerangka akuisisi teknologi MERLIN.

Elemen Kunci:

- Acquisition Strategy: Build vs Buy vs Partner. Decision framework per technology.
- Build: Core IP, competitive advantage (AI model, biotech process, MERLIN OS).
- Buy: Commodity (ERP, CRM, Office), best-of-breed, faster time-to-market.
- Partner: Strategic tech (cloud, blockchain), co-development, risk sharing.
- Vendor Evaluation: Technical fit, cost, support, security, ESG, references, financial stability.
- Negotiation: Volume discount, multi-year, exit clause, IP ownership, SLA, penalty.
- Integration: API-first, middleware, ESB, event-driven, anti-corruption layer.

Owner: CTO + Procurement MERLIN

PGA-124: Innovation Pipeline Management

Tujuan: *Manajemen pipeline inovasi MERLIN.*

Elemen Kunci:

- Pipeline Stages: Idea (200+/year) → Concept (50) → Feasibility (25) → Pilot (15) → Launch (10) → Scale (5) → Mature.
- Stage-Gate: Go/No-Go decision per stage with criteria (technical, market, financial, ESG).
- Innovation Sources: Internal R&D (60%), University (20%), Open innovation (10%), Member (5%), M&A (5%).
- Innovation Themes: Marine biotech, Blue carbon, Circular economy, Digital aquaculture, Functional food, Sustainable packaging, Bioenergy.
- Innovation Budget: 5% revenue = Rp25-42 miliar/year. Allocate per stage.
- KPI: 10 new products/year, 50% revenue from product <5 years, 1000+ patent, 50+ publication, ROI >30%.
- MERLIN Ventures: Corporate venture capital Rp100 miliar/year, invest in startup (seaweed, biotech, climate tech).

Owner: *Chief Innovation Officer MERLIN*

◆ Akhir Domain 15 — 8 Dokumen PGA ◆

DOMAIN 16 | MEDIA, NARRATIVE & COMMUNICATIONS POWER

Media, Narasi & Kekuatan Komunikasi MERLIN

Range: PGA-125 s/d PGA-132 | Total: 8 Dokumen

PGA-125: Media Relations Strategy

Tujuan: Strategi hubungan media MERLIN.

Elemen Kunci:

- Media Target: International (Bloomberg, Reuters, FT, Nikkei, SCMP), National (Kompas, Tempo, CNN Indonesia, CNBC Indonesia, Detik), Trade (SeafoodSource, FoodNavigator, NutraIngredients).
- Media Database: 500+ journalist, beat, contact, preference, history.
- Press Release: 2-4/month, distributed via PR Newswire, Bloomberg, Reuters, Antara.
- Press Conference: Quarterly, annual result, major announcement, crisis.
- Media Visit: Site visit to EBIP for journalist, annual media gathering.
- Spokesperson: CEO, CCO, topic expert. Media training annual.

Owner: Chief Communications Officer MERLIN

PGA-126: Internal Communications Master Plan

Tujuan: Master plan komunikasi internal MERLIN.

Elemen Kunci:

- Audience: 5 juta workforce (1,5 juta petani + 3,5 juta lainnya) + 10 juta family.
- Channels: Mobile app (primary), Intranet, Email, SMS, Town hall, Magazine (MERLIN Quarterly), Radio MERLIN, Notice board.
- Content: Strategy update, operational news, member success story, training, ESG, recognition, FAQ.
- Cadence: Daily (push notification), Weekly (newsletter), Monthly (magazine), Quarterly (town hall), Annual (report).
- Two-Way: Suggestion box, Q&A forum, polling, survey, focus group.
- Effectiveness: Open rate 70%+, engagement 30%+, satisfaction 80%+.

Owner: Internal Comms Director MERLIN

PGA-127: Social Media & Digital Presence

Tujuan: Social media dan kehadiran digital MERLIN.

Elemen Kunci:

- Platform Strategy: LinkedIn (B2B, 1 juta followers), Instagram (B2C, 5 juta), TikTok (youth, 3 juta), YouTube (edutainment, 2 juta subs), Twitter/X (public, 1 juta), Facebook (community, 2 juta).
- Content Pillar: Education (seaweed 101), Inspiration (member success), Product (showcase), ESG (impact), Behind-the-scene, Live.
- Posting Cadence: LinkedIn daily, Instagram 2/day, TikTok daily, YouTube weekly, Twitter 3/day.
- Influencer: 100 KOL partnership (marine, food, sustainability, business), 10 brand ambassador.
- Community: 1000+ community group, moderation, UGC, hashtag campaign.
- Advertising: Rp50 miliar/year, targeted, performance marketing, brand campaign.

Owner: Digital Marketing Director MERLIN

PGA-128: Reputation Management Framework

Tujuan: Kerangka manajemen reputasi MERLIN.

Elemen Kunci:

- Reputation Pillar: Product quality, ESG, Member welfare, Leadership, Innovation, Financial performance, Crisis response.
- Reputation Monitoring: Brand tracker (monthly), sentiment analysis (real-time), media monitoring, social listening.
- Stakeholder Perception: Survey tahunan per stakeholder (member, customer, investor, government, public, media).
- Reputation Score: Composite index 0-100. Target 80+.
- Reputation Building: Storytelling, thought leadership, ESG report, award, partnership, CSR.
- Reputation Defense: Crisis response, fact-check, counter-narrative, legal action against defamation.

Owner: Chief Reputation Officer MERLIN

PGA-129: Content Strategy & Production

Tujuan: Strategi dan produksi konten MERLIN.

Elemen Kunci:

- Content Hub: MERLIN Studios (Jakarta + Bali), 50 staff (writer, designer, videographer, editor, animator).
- Content Type: Article (blog, whitepaper), Visual (infographic, photo), Video (corporate, documentary, tutorial, short-form), Audio (podcast), Interactive (web, app, game).
- Content Calendar: Annual editorial calendar, monthly theme, weekly topic, daily post.
- Content Pillar: Education (40%), Inspiration (25%), Product (20%), ESG (10%), Corporate (5%).
- Distribution: Owned (website, app, social), Earned (PR, UGC), Paid (advertising), Shared (partner).
- Performance: Reach, engagement, conversion, sentiment, ROI. Monthly report.

Owner: Chief Content Officer MERLIN

PGA-130: Thought Leadership Program

Tujuan: Program thought leadership MERLIN.

Elemen Kunci:

- Thought Leader: CEO, C-level, Dewan Pakar. 20 spokesperson with depth expertise.
- Topic: Maritime economy, Blue carbon, Hilirisasi, Sovereign economy, Marine biotech, Sustainable tourism, ESG.
- Channel: Op-ed (Kompas, Tempo, FT, Nikkei), Speaking (WEF, APEC, B20, Indonesia ESG), Webinar, Podcast, Book, Academic journal.
- MERLIN Forum: Annual flagship forum, 1000 attendee, international speaker, broadcast.
- Publication: Whitepaper quarterly, MERLIN Journal (academic, peer-reviewed), book annual.
- KPI: 100 op-ed/year, 50 speaking/year, 5 book/year, top 3 maritime thought leader in Asia.

Owner: Chief Thought Leadership Officer MERLIN

PGA-131: Political Communication Framework

Tujuan: Kerangka komunikasi politik MERLIN.

Elemen Kunci:

- Political Neutrality: MERLIN non-partai, tidak mendukung kandidat, fokus kebijakan.
- Policy Advocacy: UU Rumpun Laut, blue economy policy, hilirisasi incentive, ESG regulation.
- Stakeholder Engagement: DPR (11 komisi), 12 kementerian, 34 provinsi, DPRD 350 kabupaten.
- Political Communication: Position paper, hearing, public hearing, coalition building, media campaign.
- Election Cycle: Pre-election (engagement all candidate), post-election (advocacy to winner).
- Compliance: No political contribution, no violation of campaign finance law, transparency.

Owner: Chief Public Affairs Officer MERLIN

PGA-132: Narrative Warfare Defense

Tujuan: Pertahanan perang narasi MERLIN.

Elemen Kunci:

- Narrative Threat: Disinformation (fake news), Misinformation (false info), Negative campaign (competitor, anti), Foreign interference.
- Monitoring: 24/7 social listening, fact-check partnership (Mafindo, CekFakta), sentiment alert.
- Response: Quick fact-check (4 hour), counter-narrative (24 hour), legal action (defamation), platform takedown.
- Counter-Narrative: Positive storytelling, evidence-based, KOL amplification, member mobilization.
- Prebunking: Pre-emptive education on common misinformation, inoculation theory.
- War Room: During high-risk event (crisis, election), 24/7 narrative command center.

Owner: Chief Narrative Defense Officer MERLIN

◆ Akhir Domain 16 — 8 Dokumen PGA ◆

DOMAIN 17 | EDUCATION, RESEARCH & KNOWLEDGE PRODUCTION

Pendidikan, Riset & Produksi Pengetahuan MERLIN

Range: PGA-133 s/d PGA-140 | Total: 8 Dokumen

PGA-133: Education System Master Plan

Tujuan: Master plan sistem pendidikan MERLIN.

Elemen Kunci:

- Akademi MERLIN: 5 kampus utama (Jakarta, Surabaya, Makassar, Bali, Sorong), 40 training center per EBIP, e-learning platform.
- Curriculum: Budidaya (3 bulan cert), Processing (6 bulan), Quality (3 bulan), Leadership (12 bulan), Marine biotech (24 bulan MSc), Maritime civilization (36 bulan PhD).
- Faculty: 1000 internal trainer, 500 eksternal expert, 50+ universitas partnership.
- Student: 100.000 active student 2030, 1,5 juta alumni (petani trained), 10.000 professional certified.
- Facility: Smart classroom, lab, workshop, library, dormitory, simulation center.
- Partnership: Dual degree dengan 10 universitas top (IPB, UI, UGM, ITS, UNHAS, Wageningen, Tokyo, NTU, ANU, Bristol).

Owner: Chief Education Officer MERLIN

PGA-134: Research & Development Framework

Tujuan: Kerangka riset dan pengembangan MERLIN.

Elemen Kunci:

- R&D Themes: Budidaya (bibit unggul, disease, IMTA), Processing (efficient extraction, novel product), Product (100+ produk), ESG (carbon, biodiversity), Tech (AI, IoT, blockchain).
- R&D Budget: 5% revenue = Rp25-42 miliar/year 2030.
- R&D Lab: MERLIN Research Center (Jakarta hub), 40 EBIP lab, partnership dengan 50 universitas, BRIN.
- Researcher: 1000+ peneliti (500 internal, 500 partner), 100 PhD, 200 MSc.
- Publication: 200+ paper/year (50% Q1 journal), 1000+ patent, 50+ book chapter.
- R&D Partnership: 50+ universitas, 10+ international lab, 20+ industry partner.

Owner: Chief R&D Officer MERLIN

PGA-135: Academy & Learning Institution

Tujuan: Akademi dan institusi pendidikan MERLIN.

Elemen Kunci:

- MERLIN University: Future full university 2035, accreditation A, 10 fakultas (Budidaya, Processing, Biotech, Maritime, Tourism, Carbon, Business, Engineering, Health, Humanities).
- MERLIN Vocational: SMK Bahari MERLIN (5 sekolah), D3 Aquaculture (5 akademi).
- MERLIN Professional: Executive education, certification, professional development.
- MERLIN Online: MOOC (massive open online course), 100 kursus, 1 juta learner.
- MERLIN Pre-School: Early childhood untuk anak member, 1000 sekolah.
- MERLIN Library: Digital library, 100.000 koleksi, akses member.

Owner: Rector MERLIN University

PGA-136: Knowledge Production & Publishing

Tujuan: Produksi pengetahuan dan penerbitan MERLIN.

Elemen Kunci:

- MERLIN Press: Penerbit buku, jurnal, laporan. 50 buku/year, 4 jurnal/year, 100 laporan/year.
- MERLIN Journal: Peer-reviewed academic journal (Indonesia + English), Q1 target, open access.
- MERLIN Magazine: Quarterly member magazine, 1,5 juta cetak, bilingual.
- MERLIN TV: YouTube channel + future OTT, documentary, edutainment, 1 juta subs.
- MERLIN Radio: Weekly program, partnership dengan RRI, coastal community reach.
- MERLIN Database: Seaweed knowledge database, public access, search engine.

Owner: Chief Knowledge Officer MERLIN

PGA-137: Think Tank & Policy Research

Tujuan: Think tank dan riset kebijakan MERLIN.

Elemen Kunci:

- MERLIN Institute: Think tank untuk maritime economy policy. Independent, credible.
- Research Agenda: 20 policy paper/year (UU Rumput Laut, blue economy, carbon market, hilirisasi, ESG).
- Fellow: 20 fellow (akademisi, praktisi, regulator), 5 senior fellow.
- Publication: Policy brief, working paper, book, op-ed. Distribute to DPR, kementerian, akademisi.
- Event: Annual MERLIN Policy Forum, monthly discussion, expert briefing.
- Advocacy: Policy advocacy to government, coalition building, public campaign.

Owner: Director MERLIN Institute

PGA-138: Intellectual Property Strategy

Tujuan: Strategi hak kekayaan intelektual MERLIN.

Elemen Kunci:

- IP Portfolio: 1000+ patent (process, product), 500+ trademark (MERLIN + sub-brand), 100+ design, 50+ copyright, 20+ trade secret.
- Patent Strategy: File early, file broad, file global (PCT). Protection 20 tahun.
- Trademark: MERLIN registered in 100+ country, defensive registration, anti-infringement.
- Trade Secret: Confidential process, NDA, access control, exit non-compete.
- Licensing: IP licensing for revenue, royalty 5-15%, territory-based, term-limited.
- Enforcement: Monitoring infringement, cease & desist, litigation, customs recordal.
- Open IP: Non-strategic IP open-sourced for community benefit, strategic IP proprietary.

Owner: Chief IP Officer MERLIN

PGA-139: Academic Partnerships & Collaborations

Tujuan: Kemitraan akademik dan kolaborasi MERLIN.

Elemen Kunci:

- Domestic Partner: IPB, UI, UGM, ITS, UNHAS, UNDIP, UNAIR, UNRI, UNPATTI, Unsrat - 50+ universitas.
- International: Wageningen, Tokyo, NTU, ANU, Bristol, UCSD, MIT - 20+ top university.
- Collaboration Models: Joint research, joint degree, student exchange, faculty exchange, joint lab, joint center.
- Industry-Academia: R&D collaboration, internship, scholarship, adjunct professor, advisory board.
- Government-Academia: BRIN, KKP, Kemenristekdikti - joint program.
- Investment: Rp50 miliar/year for partnership, 500 scholarship, 100 joint research.

Owner: Chief Academic Partnership Officer MERLIN

PGA-140: Library & Information Center

Tujuan: Perpustakaan dan pusat informasi MERLIN.

Elemen Kunci:

- Physical Library: 5 perpustakaan utama (Jakarta, Surabaya, Makassar, Bali, Sorong) + 40 mini library per EBIP.
- Digital Library: 100.000 koleksi digital (ebook, journal, report, video, audio), search engine, mobile app.
- Archive: MERLIN archive (sejarah, foto, video, dokumen), 100-year preservation.
- Museum: MERLIN Maritime Museum di Bali, sejarah rumput laut Indonesia, edutainment.
- Information Service: Reference service, research help, inter-library loan.
- Open Access: Open access for member, public access for non-strategic content.

Owner: Chief Librarian MERLIN

◆ Akhir Domain 17 — 8 Dokumen PGA ◆

DOMAIN 18 | HEALTHCARE, WELLNESS & SOCIAL SERVICES

Kesehatan, Kesejahteraan & Layanan Sosial MERLIN

Range: PGA-141 s/d PGA-148 | Total: 8 Dokumen

PGA-141: Healthcare System Design

Tujuan: *Desain sistem kesehatan MERLIN.*

Elemen Kunci:

- Healthcare Vision: Universal health coverage untuk 1,5 juta member + 10 juta family.
- MERLIN Clinic: 350 klinik (1 per kabupaten), 40 rumah sakit (1 per EBIP), telemedicine 24/7.
- Service: Primary care, emergency, specialist (internal, obgyn, pediatric, surgery, eye, dental), mental health, traditional medicine.
- Insurance: MERLIN Health Insurance (BPJS Kesehatan + private top-up), 100% covered.
- Pharmacy: MERLIN Pharmacy (350 outlet), BPJS + private, rumput laut nutraceutical.
- Investment: Rp10 triliun 2026-2030, partnership dengan Kemenkes, IDI.

Owner: *Chief Health Officer MERLIN*

PGA-142: Public Health Framework

Tujuan: *Kerangka kesehatan masyarakat MERLIN.*

Elemen Kunci:

- Public Health Priority: Maternal-child, elderly, occupational, environmental, mental, communicable, non-communicable.
- Programs: Immunization (100% child), nutrition (fortified seaweed food), sanitation (clean water 100%), mental health (screening), occupational (aquaculture safety).
- Surveillance: Disease surveillance, outbreak response, health informatics.
- Health Promotion: Healthy lifestyle, seaweed nutrition, exercise, no smoking, no alcohol.
- Environmental Health: Water quality, air quality, food safety, vector control.
- Success Metrics: Life expectancy 75+, infant mortality <15/1000, maternal mortality <100/100k, 0 outbreak.

Owner: *Chief Public Health Officer MERLIN*

PGA-143: Mental Health & Wellbeing

Tujuan: *Kesehatan mental dan kesejahteraan MERLIN.*

Elemen Kunci:

- Mental Health Service: Counseling, psychotherapy, psychiatry, support group. Stigma reduction.

- Workplace Wellbeing: Work-life balance, stress management, EAP (Employee Assistance Program), mental health day.
- Member Wellbeing: Petani stress (climate, financial, social), counseling, financial advice, community support.
- Youth Mental Health: School counselor, peer support, anti-bullying, suicide prevention.
- Elderly Wellbeing: Social activity, cognitive health, depression screening, end-of-life care.
- Crisis: Mental health crisis response, hotline 24/7, emergency psychiatric.

Owner: Chief Mental Health Officer MERLIN

PGA-144: Social Safety Net

Tujuan: Jaring pengaman sosial MERLIN.

Elemen Kunci:

- MERLIN Safety Net: Member guaranteed basic income, healthcare, housing, education, food.
- Programs: Unemployment benefit, disability benefit, elderly pension, child benefit, widow benefit, disaster relief.
- Funding: 5% revenue + member contribution + government co-funding.
- Targeting: Universal for member, with additional support for vulnerable (poor, disabled, elderly, children).
- Administration: MERLIN Social Agency, digital delivery, fraud prevention.
- Impact: 0 extreme poverty, 0 hunger, 0 homeless among member.

Owner: Chief Social Welfare Officer MERLIN

PGA-145: Housing & Shelter Policy

Tujuan: Kebijakan perumahan dan penampungan MERLIN.

Elemen Kunci:

- MERLIN Housing: 1,5 juta rumah layak untuk member petani. Subsidized, green, earthquake-resistant.
- Construction: 100.000 rumah/year 2026-2030. Prefab + local material + solar panel.
- Financing: Subsidized mortgage (0% down, 3% interest, 20 year), government subsidy, CSR.
- Disaster Shelter: 40 EBIP shelter, 350 koperasi shelter, deployable tent, container house.
- Elderly Housing: MERLIN Senior Living (5 pilot), assisted living, nursing home.
- Standard: 36 m2 minimum, 2 bedroom, kitchen, bathroom, ventilation, sanitation.

Owner: Chief Housing Officer MERLIN

PGA-146: Nutrition & Food Security

Tujuan: Nutrisi dan ketahanan pangan MERLIN.

Elemen Kunci:

- Nutrition Vision: Zero hunger, zero malnutrition, healthy diet for all member.
- Seaweed Nutrition: Rumput laut kaya iodine, fiber, vitamin, mineral. Fortifikasi pangan.
- Programs: School feeding (seaweed-fortified), maternal nutrition, elderly nutrition, food bank.
- Food Security: Domestic production, stockpile, distribution, price stabilization.
- Food Safety: Halal, organic, contaminant-free, traceable.
- Success Metrics: 0 stunting, 0 malnutrition, 100% food secure, dietary diversity score 5+.

Owner: Chief Nutrition Officer MERLIN

PGA-147: Elder Care & Aging

Tujuan: Lansia perawatan dan penuaan MERLIN.

Elemen Kunci:

- Elderly Demographic: 10% member >60 tahun by 2030, growing.
- Programs: Elderly pension (Rp2 juta/month), elderly healthcare (free), elderly housing, elderly activity center, elderly home care.
- MERLIN Senior Living: 5 pilot (Jakarta, Bali, Surabaya, Makassar, Sorong). Independent + assisted + nursing.
- Elderly Engagement: Wisdom council, mentoring, volunteering, cultural transmission.
- Elderly Protection: Anti-abuse, anti-neglect, anti-fraud, legal protection.
- End-of-Life: Palliative care, hospice, funeral support, grief counseling.

Owner: Chief Elder Care Officer MERLIN

PGA-148: Sports, Recreation & Physical Wellness

Tujuan: Olahraga, rekreasi & kesehatan fisik MERLIN.

Elemen Kunci:

- Sports Facility: 40 sports center per EBIP, 350 sports field per kabupaten, 5 national sports complex.
- Programs: Mass sport (jogging, cycling, swimming), competitive sport (MERLIN League), water sport (sailing, diving, surfing), traditional sport.
- MERLIN Games: Annual national sport event, 10.000 athlete, 20 sport.
- Recreation: 40 Blue Tourism destination, community park, cultural center, library.
- Physical Wellness: Annual health check, fitness program, nutrition, ergonomic workplace.
- Success Metrics: 70% member active sport, 50% fitness level good, 0 obesity epidemic.

Owner: Chief Sports Officer MERLIN

◆ Akhir Domain 18 — 8 Dokumen PGA ◆

DOMAIN 19 | INFRASTRUCTURE, URBAN PLANNING & DEVELOPMENT

Infrastruktur, Perencanaan Kota & Pembangunan MERLIN

Range: PGA-149 s/d PGA-156 | Total: 8 Dokumen

PGA-149: Infrastructure Master Plan

Tujuan: *Master plan infrastruktur MERLIN.*

Elemen Kunci:

- Infrastructure Portfolio: 40 EBIP, 40 Blue Tourism, 350 koperasi office, 1000 km road, 40 port, 10 airport, 5000 km fiber optic, 5 GW renewable energy.
- Investment: Rp340 triliun (50% total capex). PPP + government + own + investor.
- Phasing: 2026-2027 Phase 1 (10 EBIP), 2028 Phase 2 (20), 2029 Phase 3 (30), 2030 Phase 4 (40).
- Quality: World-class, ISO standard, green building, smart city.
- Maintenance: Asset management system, preventive maintenance, 5% asset value/year.
- Sustainability: 100% renewable, zero waste, biodiversity net positive, LEED Platinum.

Owner: *Chief Infrastructure Officer MERLIN*

PGA-150: Urban Planning & Development Framework

Tujuan: *Kerangka perencanaan dan pengembangan kota MERLIN.*

Elemen Kunci:

- MERLIN Eco-City: 40 new city per EBIP, 50.000 population each. Smart, green, livable.
- Urban Design: Mixed-use, walkable, transit-oriented, green space 30%, blue infrastructure.
- Zoning: Industrial, residential, commercial, recreational, institutional, green.
- Smart City: IoT, AI, digital twin, citizen app, e-government, intelligent transport.
- Affordable Housing: 30% housing affordable for low-income, inclusionary zoning.
- Resilience: Climate adaptation, disaster resistant, sponge city, redundancy.

Owner: *Chief Urban Planning Officer MERLIN*

PGA-151: Transportation Network

Tujuan: *Jaringan transportasi MERLIN.*

Elemen Kunci:

- Maritime: 40 port (1 per EBIP), ferry network, cargo ship, cruise terminal.
- Road: 1000 km new road connecting EBIP, highway, bridge, tunnel.
- Rail: 500 km rail for cargo (rumput laut to port), future passenger rail.

- Air: 10 airport (1 per regional), cargo terminal for export, future international.
- Urban Transit: BRT, future LRT/MRT in MERLIN Eco-City, micro-mobility.
- Logistics: Smart warehouse, cold chain, dry port, intermodal.
- Sustainability: Electric vehicle, bioethanol E30, biogas, low carbon logistics.

Owner: Chief Transportation Officer MERLIN

PGA-152: Energy Grid & Utilities

Tujuan: Jaringan energi dan utilitas MERLIN.

Elemen Kunci:

- Energy Mix: 100% renewable - biogas (40 MW from EBIP), solar (200 MW), green hydrogen (pilot), wind (potential).
- Grid: Microgrid per EBIP, smart grid, energy storage, demand response.
- Self-Sufficient: EBIP 100% self-sufficient energy, surplus to community.
- Water: Desalination (seawater), water recycle 90%, rainwater harvesting.
- Waste: Zero waste, waste-to-energy, composting, recycling.
- Telecom: Fiber optic 5000 km, 5G, satellite backup.

Owner: Chief Energy Officer MERLIN

PGA-153: Digital Infrastructure & Connectivity

Tujuan: Infrastruktur digital dan konektivitas MERLIN.

Elemen Kunci:

- Backbone: 5000 km fiber optic, 40 EBIP connected, 350 koperasi connected.
- Last Mile: 4G/5G per EBIP, satellite for remote, WiFi mesh for community.
- Data Center: MERLIN Data Center (Jakarta, Surabaya), tier III, 5 MW, sovereign cloud.
- Edge: 40 edge server per EBIP, real-time IoT processing.
- IoT Network: LoRaWAN, NB-IoT, satellite IoT for ocean monitoring.
- Cyber: DDoS protection, firewall, IDS/IPS, SOC 24/7.

Owner: Chief Digital Infrastructure Officer MERLIN

PGA-154: Water, Sanitation & Hygiene (WASH)

Tujuan: Air, sanitasi dan kebersihan MERLIN.

Elemen Kunci:

- Water Supply: 100% member access clean water, desalination, rainwater harvesting, water treatment.
- Sanitation: 100% member access toilet, sewage treatment, septic, centralized WWTP per EBIP.
- Hygiene: Hand washing, soap, education, campaign.
- Wastewater: Treatment 100%, no discharge to sea, recycle for irrigation.

- Solid Waste: 3R (reduce, reuse, recycle), zero waste to landfill, composting.
- Stormwater: Drainage, retention, sponge city.

Owner: Chief WASH Officer MERLIN

PGA-155: Waste Management & Environmental Services

Tujuan: Manajemen limbah dan layanan lingkungan MERLIN.

Elemen Kunci:

- Zero Waste: 0% to landfill, 95% resource utilization, 5% safe disposal.
- Waste Hierarchy: Reduce → Reuse → Recycle → Recover → Dispose.
- Industrial Symbiosis: Byproduct from one EBIP = input for another. Sludge → fertilizer, wash water → biogas.
- Hazardous Waste: Special handling, licensed disposal, manifest tracking.
- Marine Litter: 0 plastic discharge, bioplastic substitution, beach cleanup.
- Environmental Monitoring: Air, water, soil, biodiversity. Real-time dashboard.

Owner: Chief Environmental Services Officer MERLIN

PGA-156: Public Facilities & Civic Infrastructure

Tujuan: Fasilitas publik dan infrastruktur civic MERLIN.

Elemen Kunci:

- Education Facility: 1000 pre-school, 1000 school, 5 university, 40 training center.
- Health Facility: 350 clinic, 40 hospital, 40 elder care, 40 disability center.
- Cultural Facility: 40 museum, 40 library, 40 theater, 40 community center.
- Recreational Facility: 40 sports complex, 40 park, 40 beach club.
- Religious Facility: Mosque, church, temple, vihara per MERLIN Eco-City.
- Civic Facility: 40 city hall, 40 post office, 40 police, 40 fire station.

Owner: Chief Public Facilities Officer MERLIN

◆ Akhir Domain 19 — 8 Dokumen PGA ◆

DOMAIN 20 | LEGAL, CONSTITUTIONAL & JUSTICE SYSTEM

Sistem Hukum, Konstitusional & Keadilan MERLIN

Range: PGA-157 s/d PGA-165 (9 dokumen) | Total: 9 Dokumen

PGA-157: Legal System Design

Tujuan: *Desain sistem hukum MERLIN.*

Elemen Kunci:

- Legal Framework: UU Rumput Laut (draft), AD/ART MERLIN, bylaws 10 asosiasi, kontrak standard.
- Legal Domain: Corporate, Cooperative, Contract, Labor, Environmental, IP, Tax, Maritime.
- Legal Source: National law, Islamic law (for syariah), customary law (for adat), international treaty.
- Legal Risk: 50+ legal risk identified, mitigation, monitoring.
- Legal Counsel: 50 in-house lawyer, 10 external firm, 5 international firm.
- Legal Training: Annual legal training for staff, member legal literacy program.

Owner: *Chief Legal Officer MERLIN*

PGA-158: Judicial Administration

Tujuan: *Administrasi peradilan MERLIN.*

Elemen Kunci:

- Internal Justice: MERLIN Court (arbitration internal), per adat + per koperasi.
- External Justice: Pengadilan Negeri, Pengadilan Niaga, Pengadilan Tata Usaha Negara, MA.
- Alternative Dispute Resolution (ADR): Mediation, arbitration, expert determination.
- Enforcement: Eksekusi putusan, sita, sita eksekusi.
- Case Management: Legal case tracker, status, deadline, lawyer assignment.
- Budget: Rp50 miliar/year legal affairs, 100 active case average.

Owner: *Chief Legal Affairs Officer MERLIN*

PGA-159: Law Enforcement & Security

Tujuan: *Penegakan hukum dan keamanan MERLIN.*

Elemen Kunci:

- MERLIN Security: 10.000 personel security, 100 patrol boat, drone, CCTV, AI surveillance.
- Partnership: TNI AL, Polairud, Bakamla, KKP, Polri, BIN. Civil-military cooperation.
- Crime Prevention: Anti-theft (rumput laut), anti-illegal fishing, anti-fraud, anti-corruption.
- Investigation: Internal investigation, evidence collection, cooperation with authority.
- Prosecution: Cooperation with kejaksaan, victim support, witness protection.

- Compliance: 100% compliance with law, anti-bribery, anti-corruption, ISO 37001.

Owner: Chief Security Officer MERLIN

PGA-160: Constitutional Framework & Interpretation

Tujuan: Kerangka konstitusional dan interpretasi MERLIN.

Elemen Kunci:

- MERLIN Constitution: AD/ART + bylaws = constitution MERLIN. Highest legal document.
- Constitutional Principles: Sovereignty (rakyat member), Sustainability (zero waste), Justice (harga jaminan), Solidarity (gotong royong), Wisdom (ilmu pengetahuan).
- Interpretation: Dewan Pengawas sebagai interpreter, preseden tertulis, kasus hukum.
- Amendment: Hanya melalui Musda 2/3 member + approval Dewan Pembina + MK (if legal).
- Constitutional Review: Annual review, gap analysis, amendment proposal.
- International Alignment: UN Global Compact, ICA, ILO, UNCLOS, SDG.

Owner: Dewan Pengawas + Legal MERLIN

PGA-161: Human Rights Commission

Tujuan: Komisi hak asasi manusia MERLIN.

Elemen Kunci:

- MERLIN HRC: Independent human rights commission, 7 member, term 5 year.
- Mandate: Monitor, investigate, report, recommend on human rights in MERLIN.
- Rights: Member rights (16 hak), labor rights, women rights, child rights, indigenous rights, disability rights, environmental rights.
- Procedure: Complaint → Investigation → Report → Recommendation → Follow-up.
- Annual Report: Public annual report, transparent, submitted to Dewan Pembina + Komnas HAM.
- Independence: Budget independent, no interference, whistleblower protection.

Owner: Ketua Komisi HAM MERLIN

PGA-162: Ombudsman & Grievance Mechanism

Tujuan: Ombudsman dan mekanisme pengaduan MERLIN.

Elemen Kunci:

- MERLIN Ombudsman: Independent, 5 ombudsman, term 4 year. Receive & investigate complaint.
- Grievance Channels: Hotline (24/7), email, mobile app, walk-in, letter, social media.
- Process: Receive → Triage → Investigate → Mediate → Recommend → Follow-up. SLA per case.
- Cases: Service complaint, staff misconduct, policy issue, member dispute.
- Resolution: Mediation (70% case), investigation (20%), formal proceeding (10%).

- Transparency: Quarterly report, anonymized case published, trend analysis.

Owner: Ketua Ombudsman MERLIN

PGA-163: Anti-Corruption & Integrity Framework

Tujuan: Kerangka anti-korupsi dan integritas MERLIN.

Elemen Kunci:

- Integrity Vision: 0 corruption in MERLIN. ISO 37001 (anti-bribery) certified.
- Risk: Bribery, kickback, embezzlement, fraud, conflict of interest, nepotism.
- Prevention: Code of conduct, training, due diligence, segregation of duty, audit, whistleblower.
- Detection: Internal audit, external audit, whistleblower, data analytics, fraud indicator.
- Response: Investigation, disciplinary action, legal action, recovery, public disclosure.
- Culture: Integrity champion, recognition, transparency, tone from top.

Owner: Chief Integrity Officer MERLIN

PGA-164: Dispute Resolution & Arbitration System

Tujuan: Sistem penyelesaian sengketa dan arbitrase MERLIN.

Elemen Kunci:

- Dispute Types: Member-staff, member-member, member-koperasi, koperasi-EBIP, MERLIN-partner, MERLIN-government, MERLIN-competitor.
- Resolution Path: Negotiation → Mediation → Arbitration → Litigation. Tiered.
- MERLIN Arbitration: Internal arbitration (BANI-aligned), 7 arbitrator, fast track 3 month.
- External Arbitration: BANI (domestic), ICC/ SIAC/ HKIAC (international).
- Litigation: Pengadilan Negeri/Niaga, MA. Last resort.
- Enforcement: New York Convention 1958 for international award, eksekusi for domestic.

Owner: Chief Dispute Resolution Officer MERLIN

PGA-165: Master Compliance & Regulatory Atlas

Tujuan: Atlas master kepatuhan dan regulasi MERLIN.

Elemen Kunci:

- Comprehensive Regulatory Map: 200+ regulation (national + international) applicable to MERLIN.
- Compliance Domain: Corporate, Cooperative, Maritime, Environmental, Labor, Tax, IP, Data, Halal, International Trade, ESG, Health & Safety, Anti-Bribery.
- Compliance Dashboard: Real-time status per regulation, owner, deadline, action.
- Compliance Calendar: Reporting deadline, audit, certification renewal, regulatory change.
- Regulatory Change Management: Monitor → Impact assess → Adapt → Implement → Verify.
- Audit Trail: All compliance activity logged, evidence retained 10 year, audit ready.
- Continuous Improvement: Annual compliance review, gap analysis, remediation, certification.

- Conclusion: This Master Document 165 PGA Completed represents the most comprehensive, integrated, and sovereign-grade documentation ever created for an Indonesian maritime ecosystem. MERLIN 2030 stands as the blueprint for transforming Indonesia into the world's seaweed industrial superpower through unity of 10 associations, 40 Eco Blue Industrial Parks, 40 Blue Tourism destinations, 5 million jobs, Rp500 trillion annual export, 12 million tons CO2 sequestered, and 100+ downstream products—all built on the foundation of farmer prosperity, environmental sustainability, and national sovereignty.

Owner: Chief Compliance Officer MERLIN

◆ Akhir Domain 20 — 9 Dokumen PGA ◆

DEKLARASI PENUTUP GRAND MASTER DOCUMENT



Deklarasi MERLIN 2030

Dengan ini, Masyarakat Ekosistem Rumput Laut Indonesia (MERLIN) mendeklarasikan bahwa Master Document 165 PGA Completed ini merupakan arsitektur dokumentasi tertinggi dan paling komprehensif yang pernah disusun untuk ekosistem rumput laut Indonesia. Dokumen ini mencerminkan konvergensi kebijakan strategis, tata kelola modern, inovasi teknologi, keadilan sosial, kedaulatan ekonomi, dan visi peradaban bahari Indonesia abad ke-21.

Seluruh 165 dokumen PGA dalam 20 Domain ini disusun untuk memastikan bahwa MERLIN 2030 bukan sekadar visi, melainkan blueprint yang dapat diimplementasikan, diaudit, dan dievaluasi secara transparan. Setiap dokumen memiliki pemilik (owner) yang jelas, elemen-elemen yang terukur, dan keterkaitan dengan dokumen lainnya sehingga membentuk satu kesatuan ekosistem yang utuh dan berdaulat.

Komitmen MERLIN: (1) Menjadikan Indonesia Pusat Industri Hilirisasi Rumput Laut Dunia pada 2030; (2) Mengangkat 1,5 juta KK petani rumput laut menjadi sejahtera; (3) Mewujudkan 40 Eco Blue Industrial Park dengan standar zero waste; (4) Menghasilkan Rp500 triliun ekspor per tahun; (5) Menyerap 12 juta ton CO₂ per tahun melalui blue carbon; (6) Menciptakan 5 juta lapangan kerja baru; (7) Menjaga kedaulatan ekonomi maritim Indonesia untuk generasi kini dan mendatang.

Pengesahan

Dewan Pembina MERLIN

(Presiden Republik Indonesia)

Dewan Pakar MERLIN

(Ketua Dewan Pakar 46 Lintas
Bidang)

Ketua Umum MERLIN

(Ketua Umum MERLIN)

Jakarta, Juli 2026

**MERLIN 2030 — DARI LAUT UNTUK NUSANTARA, DARI NUSANTARA UNTUK
DUNIA**